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Equilibrium Robustness for Non-Convex Minimax Problems with Uncertain Constraints

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Abstract

Minimax problems, which model two-player zero-sum games, have gained significant attention due to recent advances in machine learning, particularly in non-convex settings. Yet, the robustness of such problems under uncertainty in both players' constraints is still insufficiently understood. This thesis investigates the probabilistic robustness of nonconvex–nonconcave (NC–NC) minimax problems with uncertain constraints via the scenario approach.

Robustness analysis is mathematically challenging due to the NC-NC structure of the payoff function and the presence of uncertain constraints. In such settings, a Nash equilibrium (NE) may fail to exist or lose desirable well-posedness properties, which significantly complicates the analysis. Thus, the robustness of equilibrium solutions remains an open question.

Building on recent developments in non-convex scenario optimization, this thesis derives probabilistic robustness guarantees to overcome these challenges. The main contributions are twofold:

- Under convex constraint sets: A tight probabilistic robustness guarantee is established for an ε -stationary point. By proving the monotonicity of the stationary residual with respect to the number of scenarios, the standard dependence on the non-degeneracy assumption is bypassed.
- Under non-convex constraint sets: The fundamental limitations of obtaining tight theoretical bounds within existing frameworks are revealed. Consequently, a relaxed, yet rigorously valid, probabilistic bound is developed for a global minimax point.

Finally, extensive numerical experiments are provided to corroborate the theoretical findings.

Keywords

Uncertainty, Data-driven robust optimization, Scenario approach, Non-convex problems, Minimax problems

Sammanfattning

Minimax-problem, vilka används för att modellera nollsummespel med två spelare, har fått stor uppmärksamhet till följd av den senaste tidens framsteg inom maskininlärning, i synnerhet för icke-konvexa problemformuleringar. Trots detta är robustheten hos sådana problem under osäkerhet i båda spelarnas bivillkor fortfarande otillräckligt förstådd. Detta examensarbete undersöker den probabilistiska robustheten hos icke-konvexa–icke-konkava (NC–NC) minimax-problem med osäkra bivillkor med hjälp av scenariometoden.

Att analysera robustheten är matematiskt utmanande på grund av utfallsfunktionens NC–NC-struktur och förekomsten av osäkra bivillkor. Under sådana förhållanden kan det hända att en Nashjämvikt (NE) inte existerar eller att den förlorar önskvärda välställdhetsegenskaper, vilket komplicerar analysen avsevärt. Följaktligen förblir jämviktslösningarnas robusthet en öppen fråga.

Med utgångspunkt i nyare framsteg inom icke-konvex scenariooptimering härleder detta examensarbete probabilistiska robusthetsgarantier för att hantera dessa svårigheter. Examensarbetet ger i huvudsak följande två bidrag:

- Givet konvexa bivillkorsmängder: En skarp probabilistisk robusthetsgaranti etableras för en ε -stationär punkt. Genom att bevisa att stationaritetsresidualen är monoton med avseende på antalet scenarier, kringgås det sedvanliga beroendet av ett icke-degenereringsantagande.
- Givet icke-konvexa bivillkorsmängder: De grundläggande begränsningarna i att härleda skarpa teoretiska skrankor inom befintliga ramverk påvisas. Som en konsekvens utvecklas en relaxerad, men matematiskt strikt, probabilistisk skranka för en global minimax-punkt.

Slutligen presenteras omfattande numeriska experiment som stödjer de teoretiska resultaten.

Nyckelord

Osäkerhet, Datadriven robust optimering, Scenariometoden, Icke-konvexa problem, Minimax-problem

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List of acronyms and abbreviations

CCP	Chance-Constrained Problem
CVaR	Conditional Value-at-Risk
DRO	Distributionally Robust Optimization
EV	Electric Vehicle
GANs	Generative Adversarial Networks
GDA	Gradient Descent Ascent
KŁ	Kurdyka-Łojasiewicz
MVI	Minty Variational Inequality
NC–NC	Nonconvex–Nonconcave
NE	Nash Equilibrium
PŁ	Polyak-Łojasiewicz
SVMs	Support Vector Machines
UAV	Unmanned Aerial Vehicle
VC	Vapnik-Chervonenkis
VI	Variational Inequalities

Chapter 1

Introduction

In many practical systems, decision-making processes are affected by various forms of uncertainty, including unpredictable environmental dynamics and incomplete information about other participants. These uncertainties complicate the analysis of strategic interactions between players. Game theory provides a natural framework for modeling such interactions by capturing the interdependence among players' decisions. Motivated by the widespread presence of uncertainty in practical applications, the study of game-theoretic problems under uncertainty has attracted increasing attention in recent years.

1.1 Motivation

Since the foundational work of von Neumann and Nash, non-cooperative game theory has become a fundamental framework for modeling strategic interactions among self-interested players. Such models have been extensively studied in the control community and applied in diverse areas, including transportation systems [1], smart grids [2], social networks [3], and mobile sensor networks [4].

A significant and widely studied class of non-cooperative games is the two-player *zero-sum game*, which provides a natural framework for modeling competitive environments. Minimax problems can be interpreted as such games, where one player seeks to minimize a payoff function while the other seeks to maximize it. This structure underpins a vast array of important applications, from optimal control [5] to machine learning [6]. For the convex case, the minimax problem is well-studied in the classic literature of von Neumann, Fan, and Sion [7, 8]. Recent advances in machine learning have spurred growing interest in non-convex minimax problems.

Prominent examples include generative adversarial networks (GANs) [9] and adversarial training [10], which can often be formulated as non-convex minimax problems.

In game-theoretic frameworks, players are typically bound by constraints that define their choice domains. Yet, in many real-world applications, such as electric vehicle (EV) charging problem and unmanned aerial vehicle (UAV) swarms, the decision-making process is inherently affected by diverse sources of uncertainty. For example, the EV charging problem is often subject to high uncertainties in renewable power generation and charging demand, as well as electricity prices [11]. In the realm of multi-robot systems, such as UAV swarms, agents must execute cooperative or competitive tasks while navigating highly uncertain environments characterized by wind disturbances, intermittent communication links, and noisy partial observations [12].

These unavoidable uncertainties make it hard to rely on traditional deterministic game-theoretic models. If such variability is ignored, the resulting strategies may perform poorly or even fail in real-world settings. This highlights the need for new game-theoretic approaches that explicitly account for uncertainty and make more robust and reliable decisions in practice.

1.2 Related Work

The literature on game theory under uncertainty has traditionally been dominated by two primary paradigms. The first is robust optimization, which relies on the predefined geometric structure of uncertainty sets. The second is stochastic optimization, which requires prior knowledge of the underlying probability distributions. To overcome the respective limitations of these classical methods, distributionally robust optimization (DRO) and the scenario approach have recently emerged as powerful alternatives. Next, we briefly review the relevant literature along these four trajectories:

Robust optimization. Utilizing robust optimization to deal with uncertainties in game theory has a long history. In such a paradigm, the geometric structure of uncertainty sets is typically assumed to be, e.g., polyhedral, box, ellipsoidal, etc. For instance, [13] investigates incomplete-information games, proposing a robust optimization framework to handle payoff uncertainty. Similarly, [14, 15] extend this methodology to generalized games subject to uncertain coupling constraints. Here, a generalized game implies that the feasible decision set of each player may depend on the strategies of the other players. However, in practical applications, obtaining precise knowledge of the uncertainty set is often unrealistic or even impossible.

Stochastic optimization. Stochastic Nash games are predominantly analyzed via stochastic variational inequalities (VIs), assuming precise probability distributions for payoffs. For instance, [16] establishes equilibrium existence under both expected and risk-averse, e.g., CVaR (conditional value-at-risk) criteria. To compute these equilibria, [17] employs distributed stochastic approximation, such as Tikhonov regularization and proximal-point methods, ensuring global convergence. However, these stochastic approaches rely heavily on exact distributional knowledge, which is often difficult to obtain in highly non-stationary environments.

Distributionally robust optimization. DRO bridges the gap between robust and stochastic paradigms by optimizing over a predefined ambiguity set rather than an exact distribution. For generalized games with soft (i.e., probabilistic) coupling constraints, [18] employs Wasserstein-based joint chance constraints. This formulation initially leads to a mixed-integer problem, which is then solved by introducing an additional “dummy” agent. Furthermore, [19] addresses risk-averse learning in online convex games, proposing zeroth- and first-order CVaR algorithms that rigorously guarantee high-probability convergence despite inherent gradient biases. Nevertheless, properly calibrating the ambiguity set to balance conservatism and robustness remains a critical open challenge in DRO.

Scenario optimization. Recently, there has been growing interest in utilizing a data-driven paradigm known as the *scenario approach* [20] to handle uncertainties in game theory. It is based on the idea that a deterministic optimization problem is solved to obtain a scenario solution by solely collecting finite samples (also called “scenarios”) from the uncertainty, instead of directly solving the inherently-difficult uncertain optimization problem. Supported by rigorous theoretical foundations established in the scenario approach, the scenario solution is robust to unseen samples with a very high probability. In other words, this solution remains feasible for almost all future realizations of the uncertainty with a very high probability. A prominent advantage of this approach lies in its *distribution-free* characteristics, i.e., it is applicable to any distribution of the uncertainty.

Most existing analyses in the scenario approach, however, critically rely on the assumption of convexity; when this assumption is violated, the resulting bounds may not be directly applied. To address this limitation, [21] first studied general non-convex scenario optimization, while [22] subsequently achieved significantly tighter risk bounds across a broad class of (non-convex) problems, albeit with more technically involved derivations. More importantly, [22] proposed a new scenario-based decision-making framework,

opening a new door for problems beyond optimization.

The robustness of minimax problems has typically been studied under limited settings. While [23] addressed the convex case and [24] extended the analysis to non-convex cases, research has predominantly focused on probabilistic robustness with uncertainty confined to the max-player's strategy set (this corresponds to a minimax reformulation of problems with uncertainty on the cost function). A critical gap remains, namely, when the strategy sets of both players are affected by uncertainty. Addressing this problem, particularly when the payoff is NC–NC with respect to the two players' strategies, is relevant and mathematically challenging. Moreover, a Nash equilibrium (NE) in this case may fail to exist or may lack standard well-posedness properties. Therefore, the robustness of equilibria under these conditions remains an open question.

While pioneering works have successfully integrated the scenario approach with game theory and variational inequalities [25], the existing literature predominantly relies on the assumption of convexity (e.g., [11, 26]). Building upon recent breakthroughs in non-convex scenario optimization [21, 22], this thesis extends the application of the scenario approach to non-convex minimax problems. The work presented in this thesis was conducted in collaboration with the supervisor and examiner, and has been documented in a preprint [27]. This work has been accepted for publication in the Proceedings of the 23rd IFAC World Congress 2026.

1.3 Contributions

The contribution of the thesis is listed as follows:

1. The robustness of an ε -stationary point, the first-order condition for a local NE, is established under NC–NC payoffs and convex constraint sets. By demonstrating that the stationary residual is non-increasing as the number of scenarios increases, a tight probabilistic robustness guarantee for the ε -stationary point is derived (Theorem 5).
2. To further investigate the case with non-convex constraint sets, a probabilistic robustness guarantee for a global minimax point is derived by explicitly accounting for the sequential order of the players' decision-making. By leveraging the extreme value theorem, the fundamental difficulty in establishing a tight bound is proven; consequently, an alternative, formally valid robustness bound is provided for this fully non-convex setting (Theorem 6).

3. Detailed numerical experiments are conducted to corroborate the theoretical results obtained in this thesis. Specifically, the application of Theorem 5 to compute an ε -stationary point is explained using a tailored algorithm that accommodates a user-defined tolerance ε . Furthermore, a systematic procedure of a zeroth-order algorithm for seeking a global minimax point is proposed through a concrete numerical example.

1.4 Structure of The Thesis

The remainder of the thesis is organized as follows:

- Chapter 2 establishes the theoretical foundations for this thesis, introducing essential concepts and intuition related to the scenario approach and minimax problems. Specifically, Section 2.1 details the development of the scenario approach, with a particular focus on recent advances in non-convex optimization. Furthermore, given the inherent difficulty and the rapid recent development of NC–NC minimax problems, a selective overview of this domain is provided in Section 2.2. This overview focuses on the definitions of key equilibria and outlines relevant algorithms under various regularity conditions.
- Chapter 3 presents the main theoretical results: Section 3.1 formulates the research problem addressed in this thesis. Section 3.2 establishes a tight probabilistic robustness guarantee for an ε -stationary point with NC–NC payoffs under uncertain convex constraints. Section 3.3 demonstrates that global minimax points adhere to an alternative, relaxed robustness bound under uncertain nonconvex constraints (i.e., NC–NC payoffs and uncertain non-convex constraints).
- Chapter 4 presents two detailed numerical examples to validate the main theoretical findings of this thesis.
- Chapter 5 concludes the thesis and outlines promising directions for future research. In particular, it points out the limitations of this thesis, and discusses the possibility of utilizing the scenario approach to analyze the probabilistic robustness guarantees for other equilibria.

Chapter 2

Preliminaries

This chapter introduces the mathematical preliminaries foundational to this thesis, including the scenario approach, alongside key equilibria and algorithmic frameworks for minimax problems.

2.1 The Scenario Approach

The scenario approach was first introduced around the mid-2000s, and largely developed by Prof. Marco C. Campi and Prof. Simone Garatti, as well as contributions from a vast community of researchers worldwide. Thanks to these researchers, the scenario approach has been successfully applied in many problems, such as robust control [28], model predictive control [29], system identification [30], machine learning [31], finance [32], and game theory [33]. A distinguishing feature of this approach is that it neither requires any knowledge of the uncertainty set, as in robust optimization, nor of its probability distribution, as in stochastic optimization. It should be noted that only selected results of the scenario approach are provided here from a mathematical perspective; comprehensive applications and their broader implications are thoroughly covered in the book [34] and survey paper [20].

2.1.1 Introduction and Motivation

Let us start from the widely-used *scenario optimization* problem. Consider optimization problems with uncertain constraints where $f(x)$ is an objective function to be minimized, and $x \in \mathcal{X}$ is the optimization variable, with $\mathcal{X}$ being thought of as any set, i.e., finite dimensional or infinite dimensional. We assume $\mathcal{X} \subseteq \mathbb{R}^d$ here for simplicity. Let δ be a random variable drawn

from a probability space $(\Delta, \mathcal{B}(\Delta), \mathbb{P})$, where the support set Δ denotes the set of possible values for the uncertainty parameter δ , while $\mathcal{B}(\Delta)$ denotes the Borel σ -algebra on Δ , and $\mathbb{P}$ is a probability measure on $\mathcal{B}(\Delta)$. Therefore, the uncertain optimization problem is

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & x \in \mathcal{X}_\delta, \delta \in \Delta \end{aligned} \quad (2.1)$$

where $\mathcal{X}_\delta$ is a family of constraint sets parameterized by uncertain parameter $\delta \in \Delta$.

We present a simple multiproduct assembly problem to ground the abstract optimization model introduced above.

Example 1. (*Multiproduct Assembly [35, Sec. 1.3]*) Consider a company that manufactures n products using m types of subassemblies ordered from suppliers. Let a_{ij} denote the number of units of subassembly j required to produce one unit of product i , where $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$. Let δ be a realization of the random demand Δ for the products. The manufacturer pre-orders the subassemblies at a unit cost of c_j for each subassembly j . Let x_j denote the quantity of subassembly j ordered. The optimization goal is to minimize the total procurement cost while ensuring that the demand is met without exceeding the available inventory for any realization of demand. This problem is formulated as follows:

$$\begin{aligned} \min_{x \geq 0} \quad & c^\top x \\ \text{s.t.} \quad & A^\top \delta \leq x, \forall \delta \in \Delta, \end{aligned}$$

where $c = \text{col}(c_1, \dots, c_m)$, $x = \text{col}(x_1, \dots, x_m)$, and a_{ij} is the i -th row and j -th column element of matrix $A \in \mathbb{R}^{n \times m}$.

Here, the constraint $A^\top \delta \leq x$ for the decision variable x can be abstracted as an uncertain constraint $\mathcal{X}_\delta = \{x : A^\top \delta \leq x\}$, which depends on the uncertain parameter (demand) δ .

Generally, one cannot expect a solution to (2.1) unless additional information on the probability distribution $\mathbb{P}$ or the support set Δ is known. Instead, the user is assumed to have finite observations (also called “scenarios”) of the random variable δ , denoted by $\boldsymbol{\delta}^N := (\delta_1, \dots, \delta_N) \in \Delta^N$, henceforward referred to as a multi-sample. Since the scenarios are drawn independently, this multi-sample is defined on the product probability space $(\Delta^N, \mathcal{B}(\Delta)^{\otimes N}, \mathbb{P}^N)$, resulting from the N -fold Cartesian product of the original

probability space $(\Delta, \mathcal{B}(\Delta), \mathbb{P})$. This data-driven framework leads to the following deterministic scenario optimization problem

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & x \in \bigcap_{i=1}^N \mathcal{X}_{\delta_i}, \end{aligned} \tag{2.2}$$

which is generally solvable under some mild assumptions. The basic idea behind the scenario approach is to solve the deterministic scenario problem (2.2) to obtain an approximate solution x_N^* (or “scenario solution”), instead of an actual solution to the original uncertain problem (2.1), and also guarantee that this scenario solution is robust to all uncertain constraints with a very high probability (i.e., very close to 1).

For a solution x_N^* to the scenario problem (2.2), a natural concern is its robustness against new and unseen samples outside the chosen multi-sample $\boldsymbol{\delta}^N$. In the context of *generalization theory*, this question is about inferring the unknown (i.e., the whole uncertainty set Δ) from the known (i.e., the multi-sample $\boldsymbol{\delta}^N$). We give some standard definitions from [22].

Definition 1. The risk $V : \mathbb{R}^d \rightarrow [0, 1]$ of a point x is defined as

$$V(x) := \mathbb{P}\{\delta \in \Delta : x \notin \mathcal{X}_\delta\}.$$

$V(x)$ quantifies the probability that a point x is infeasible for a new and randomly drawn constraint $\mathcal{X}_{\tilde{\delta}}$ with $\tilde{\delta} \in \Delta$. For a fixed x , this quantity $V(x)$ is deterministic. Crucially, the dependence of x_N^* on the multi-sample $\boldsymbol{\delta}^N$ renders the solution itself a random variable. Therefore, the risk $V(x_N^*)$ is also a random variable; the goal of the scenario approach is to analyze the statistical properties of $V(x_N^*)$, such as its distribution or expectation. Central concepts in the scenario approach also include *support list* and *complexity*, defined as follows.

Definition 2. Given a multi-sample $\boldsymbol{\delta}^N$, a support list is a sequence of elements $\delta_{i_1}, \dots, \delta_{i_k}$ of $\boldsymbol{\delta}^N$ with $i_1 < \dots < i_k$ and $k \leq N$ such that

1. removing all elements of $\boldsymbol{\delta}^N$ except those in the sequence will not change the solution x_*^N ;
2. no more elements can be removed from the sequence while leaving the solution x_*^N unchanged.

Definition 3. The complexity of a multi-sample is the minimal cardinality among all its support lists, denoted as s_N^* .

Notably, a constraint of the scenario program (2.2) is a *support constraint* if its removal changes the solution x_N^* . Support constraints are always active constraints; however, the converse is not necessary to be true. Moreover, the scenario problem (2.2) is said to be *non-degenerate* if all active constraints are support constraints. Equivalently, this happens if for any $N \in \mathbb{N}$, there exists a unique support list with probability 1 over the multi-sample δ^N . Geometrically, this implies that the boundary of distinct constraints accumulate anomalously with zero probability. The following proposition summarizes different but mathematically the same expressions about the assumption of non-degeneracy.

Proposition 1. *The following statements of non-degeneracy are equivalent:*

1. *The scenario problem (2.2) is non-degenerate;*
2. *All active constraints are support constraints;*
3. *With probability 1, the solution to (2.2) is exactly the same the solution to the program with only the support constraints;*
4. *There exists a unique support list.*

Proof. $1 \Leftrightarrow 2$ by definition. For a scenario problem, the active constraints are unique; thus, if all active constraints are support constraints, then the support list is also unique ($2 \Rightarrow 4$). Conversely, suppose there exists an active constraint which is not of support; this would lead to multiple possible support lists, establishing $2 \Leftrightarrow 4$. Finally, $3 \Leftrightarrow 4$ is already shown in [22, Remark 1]. This completes the proof. $\square$

In many cases, the optimal solution x_N^* is “supported” by only a subset of the multi-sample δ^N ; that is, x_N^* could also be obtained by solving a scenario optimization with less constraints than (2.2). Since removing the non-degeneracy assumption allows for the existence of multiple support lists, we define the complexity as the minimal cardinality among all such lists.

It is proven that convex scenario optimization problems are characterized by a bounded number of support constraints. Using Helly’s lemma, [36, Th. 2] demonstrates that $s_N^* \leq d$ for any convex scenario program.

An intuitive example to illustrate support constraints, active constraints and the non-degeneracy case is as follows.

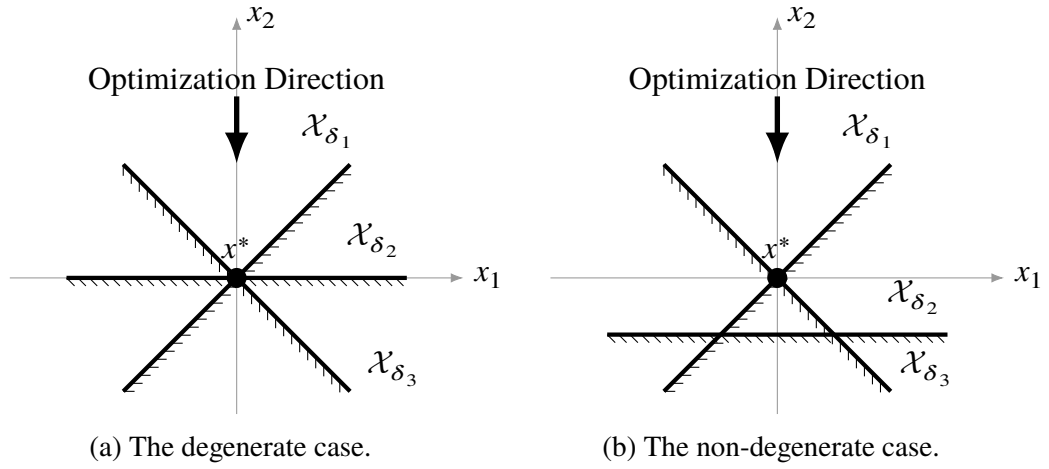


Figure 2.1: Geometric explanation of the optimization problems in Example 2. Shadings indicate the forbidden regions.

Example 2. Consider the following convex optimization problem

$$\begin{aligned} \min_{x=(x_1, x_2)} \quad & c^\top x \\ \text{s.t.} \quad & x \in \bigcap_{i=1,2,3} \mathcal{X}_{\delta_i}, \end{aligned}$$

where $c = (0, 1)$, $\mathcal{X}_{\delta_1} = \{x : x_1 - x_2 \leq 0\}$, $\mathcal{X}_{\delta_2} = \{x : x_2 \geq 0\}$, and $\mathcal{X}_{\delta_3} = \{x : x_1 + x_2 \geq 0\}$. A geometric explanation of this problem is shown in Figure 2.1a. All constraints are active constraints, but no constraint is a support constraint. Here, the three constraints coincide at the optimal point, where this problem is only 2-dimensional. Therefore, this problem is degenerate. However, if $\mathcal{X}_{\delta_2}$ becomes $\{x : x_2 \geq -0.5\}$ while the other two constraints remain unchanged, this new optimization problem is non-degenerate (see Figure 2.1b). There are two support constraints, which are exactly the two active constraints. Thus, the complexity $s_N^* = 2 \leq d = 2$.

2.1.2 Recap of Some Fundamental Results

Let us revisit several cornerstones of scenario optimization. While early contributions to the field were conceptually intuitive, recent developments have become increasingly sophisticated and abstract. We therefore feel it necessary to revisit some fundamental results. These findings are primarily published in top-tier journals such as *Mathematical Programming* and *SIAM Journal on Optimization*, as well as *IEEE Transactions on Automatic Control*.

In the very early (perhaps first) paper on scenario optimization [37], it is proven that when the scenario problem (2.2) is convex (i.e., all involved constraints and cost functions are convex), it holds that

$$\mathbb{E}_{\mathbb{P}^N}[V(x_N^*)] \leq \frac{d}{N+1}, \quad (2.3)$$

where $\mathbb{E}_{\mathbb{P}^N}[\cdot]$ denotes the expectation under probability measure $\mathbb{P}^N$. The power of this fundamental result lies in its generality for *any* convex programs. Later on, [28] first gave an explicit upper on the probability of the risk $V(x_N^*)$ exceeding a certain level ϵ , namely,

$$\mathbb{P}^N\{V(x_N^*) > \epsilon\} \leq \binom{N}{d}(1-\epsilon)^{N-d}.$$

In a significant step forward, [38] proved that the distribution of $V(x_N^*)$ is a first-order stochastically dominated Beta distribution with parameters d and $N-d+1$ (i.e., $V(x_N^*) \sim \text{Beta}(d, N-d+1)$ and we immediately obtain (2.3)), and improved the result of [28] to a very elegant and well-known result:

$$\mathbb{P}^N\{V(x_N^*) > \epsilon\} \leq \sum_{i=0}^{d-1} \binom{N}{i} \epsilon^i (1-\epsilon)^{N-i}. \quad (2.4)$$

Specifically, for problems where the number of support constraints is exactly d (called “fully-supported” in [38]), (2.4) holds with equality. In most cases, however, optimization problems are not fully-supported. Clearly, to achieve a given risk level of at most ϵ and a high confidence level $1 - \beta$, one can choose the smallest number of scenarios N such that $\sum_{i=0}^{d-1} \binom{N}{i} \epsilon^i (1-\epsilon)^{N-i} \leq \beta$. Moreover, there is a sufficient condition [34, Ch. 3.2.1] to achieve this is $N \geq \frac{2}{\epsilon} \left(\ln \frac{1}{\beta} + d - 1 \right)$. [39] later improved this sample bound in the cases when the chance constraints leave a linear subspace unconstrained, while [40] proposed incremental algorithms to minimize the number of scenarios based on the statistical connection between complexity and risk.

All results introduced so far characterize finite-sample behavior. A fundamental question regarding the scenario approach is its asymptotic properties as the number of scenarios tends to infinity. In this context, [41] formally proved the consistency* of the scenario approach, detailing the asymptotic behavior of the risk $V(x_N^*)$ as the sample size N tends to infinity. Furthermore, this analysis was extended to non-convex cases by [24].

*The term “consistency” used here should not be confused with the “consistency property” introduced later, which refers to an entirely different concept.

At the same time, [23] introduced a fundamental ordered Dirichlet distribution to scenario optimization with uncertainty on the objective functions $\tilde{f}(x, \delta)$, rather than constraints, which has the minimax form

$$\min_{x \in \mathcal{X}} \max_{i=1, \dots, N} \tilde{f}(x, \delta_i).$$

This problem is also studied in non-convex cases in [24]. Since this thesis focuses on problems with uncertain constraints, interested readers are kindly referred to [24, 23] for details about problems with uncertain objective functions.

All the results aforementioned are epistemologically *a priori*^{*}. That is, they rely solely on the problem dimension independent of observed data, meaning they can be obtained even before the multi-sample δ^N is collected. A fundamental breakthrough was made in [43], which introduced a new “wait-and-judge” perspective such that the predetermined constant ϵ , could be replaced by a function of the complexity s_N^* to yield a generally tighter bound compared to the tail of the binomial distribution in (2.4). Readers who are familiar with statistical learning theory may naturally recall the similarity between this perspective and the concepts of Vapnik-Chervonenkis (VC) dimension and support vector machines (SVMs), see, e.g., [44]. When the optimal solution x_N^* is obtained, one can count the number of support constraints (i.e., complexity)[†], and use this information to evaluate the risk. In this way, this new perspective is *a posteriori*. The result is formally stated in the following theorem.

Theorem 1 ([43, Th. 3]). *Support there exists at least one solution x_N^* to (2.2), and (2.2) is non-degenerate. Let $h(n)$ be any $[0, 1]$ -valued function for $n = 0, 1, \dots, N$. Given a multi-sample δ^N , the solution x_N^* to (2.2) satisfies*

$$\mathbb{P}^N\{V(x_N^*) > h(s_N^*)\} \leq \gamma^*,$$

^{*}Building upon the results of [38], the “sampling-and-discarding” approach was introduced in [42]. The core idea is to sample N scenarios but arbitrarily discard k of them to improve the cost function at the expense of a decreased feasibility when solving a chance-constrained problem (CCP). For a solution $x_{N,k}^*$ to the reduced CCP, and risk and confidence parameters $(\epsilon, \beta) \in (0, 1)^2$, the probabilistic guarantee $\mathbb{P}^N\{V(x_{N,k}^*)\} \leq \beta$ remains valid provided that $\binom{k+d-1}{k} \sum_{i=0}^{k+d-1} \binom{N}{i} \epsilon^i (1-\epsilon)^{N-i} \leq \beta$. Conceptually, this mechanism is somewhat akin to complexity in later scenario optimization, and compression schemes in machine learning (e.g., VC dimension, SVMs); however, because the discarding rule can be arbitrary, it remains strictly distinct from the intrinsic concept of a support list.

[†]An efficient way is to focus on active constraints only, since all support constraints are active constraints.

where s_N^* is the complexity of the multi-sample δ^N , and

$$\gamma^* = \inf_{\xi(\cdot) \in \mathbf{P}_N} \xi(1) \quad (2.5)$$

subject to

$$\frac{1}{k!} \frac{d^k}{dt^k} \xi(t) \geq \binom{N}{k} t^{N-k} \cdot \mathbb{1}\{t \in [0, 1 - \epsilon(k)]\}$$

for all $t \in [0, 1]$ and all $k = 0, 1, \dots, N$, where $\mathbb{1}\{\cdot\}$ denotes the indicator function, which equals 1 if the condition inside is true, and 0 otherwise; $\mathbf{P}_N$ denotes the class of polynomials of degree N .

By setting $h(k) = \epsilon$ for all k , the result in Theorem 1 is reduced to the tail of the binomial distribution in (2.4) (see [43, Cor. 1 & Remark 1]). In [45], it is also shown that complexity is a precise indicator of risk due to their concentrated statistical relationship. Specifically, the risk $V(x_N^*)$ is bounded by two functions of the complexity s_N^* like a “sandwich”. In [46], it demonstrates that without incorporating prior knowledge beyond the raw data, it is fundamentally impossible to establish conditional probabilistic guarantees regarding out-of-sample risk based solely on the solution’s complexity.

All the results discussed already, except [24], rely on the convexity assumption; that is, all sets and functions involved are assumed to be convex. To address this, several work studied non-convex problems, e.g., [47, 48], while a general framework was established in [21]. Importantly, a very recent breakthrough in [22] removed the non-degeneracy assumption in Theorem 1, and got a tighter bound compared to [21], which implies the result in [22] also holds in non-convex cases. Unfortunately, these results in non-convex cases are hold in an *a posteriori* manner, while *a priori* result without the non-degenerate assumption is hard to obtain, as discussed in [22, Sec. 3.2]. The breakthroughs in [22] are introduced in the context of more general decision-making problems in the next subsection.

2.1.3 Scenario-based Decision Making

How to make decisions under uncertainty? In Prof. Marco Campi’s words, just “let the data speak”. In recent years, the scenario approach has been extended to more general decision-making problems, rather than only optimization problems. Two major breakthroughs [45, 22] established a very elegant framework to make decisions based on available scenarios.

We start from an abstract and general set-up, and then provide some intuitive explanations. Let a decision z be chosen from a generic set (or

“decision set”) $\mathcal{Z}$, and let $\mathcal{Z}_\delta \subseteq \mathcal{Z}$ be a set consisting of all decisions made according to the scenario δ . For instance, in scenario optimization problems, $\mathcal{Z}_\delta$ denotes the set of optimal solutions for a given δ . The decision making process, which could be treated as algorithms to find the optimal solutions in optimization problems, is abstracted as a map from observed scenario(s) to decisions. Namely, we consider the following map^{*}

$$\mathcal{M}_n : \Delta^n \rightarrow \mathcal{Z}, \quad n = 0, 1, 2, \dots$$

Normatively, not only the observed data/samples (i.e., scenarios), the decision process/rule (i.e., map) itself also plays an important role in making decisions. Thus, the map must be “good enough” such that the decisions made by it are also “good”. [45] first proposed a property, later called the *consistency property* in [22], on the map $\mathcal{M}_n$. Intuitively, if $\mathcal{M}_n$ satisfies such consistency property, then the decisions made by it are robust with a very high probability.

Theorem 2 ([22, Th. 3]). *Given a multi-sample $\boldsymbol{\delta}^M$, if there is a map $\mathcal{M}_m$ satisfying the consistency property defined as:*

1. **Permutation-invariance:** *if $\delta_{i_1}, \dots, \delta_{i_m}$ is a permutation of $\delta_1, \dots, \delta_m$, then*

$$\mathcal{M}_m(\delta_{i_1}, \dots, \delta_{i_m}) = \mathcal{M}_m(\delta_1, \dots, \delta_m);$$

2. **Confirmation under appropriateness:** *if $z_m^* \in \mathcal{Z}_{\delta_{m+i}}$ for all $i = 1, \dots, n$, then*

$$z_m^* = \mathcal{M}_m(\delta_1, \dots, \delta_m) = \mathcal{M}_{m+n}(\delta_1, \dots, \delta_{m+n}) = z_{m+n}^*;$$

3. **Responsiveness to inappropriateness:** *if $z_m^* \notin \mathcal{Z}_{\delta_{m+i}}$ for at least one $i \in \{1, \dots, n\}$, then*

$$z_m^* = \mathcal{M}_m(\delta_1, \dots, \delta_m) \neq \mathcal{M}_{m+n}(\delta_1, \dots, \delta_{m+n}) = z_{m+n}^*;$$

then we have

$$\mathbb{P}^N\{V(z_N^*) > \epsilon(s_N^*)\} \leq \gamma^*,$$

where s_N^* is the complexity of the multi-sample $\boldsymbol{\delta}^N$, $\epsilon(k), k = 0, 1, \dots, N$ is any $[0, 1]$ -valued function, and γ^* is defined in (2.5).

^{*} n is allowed to be zero, which means this decision-making relies on no data/scenarios. For scenario optimization problems, this is mathematically equivalent to unconstrained optimization.

Notice that γ^* in (2.5) is obtained by solving a semi-infinite optimization problem, which is inherently difficult to solve. Specifically, γ^* equals to the infimum of all polynomials of degree N , $\xi(\cdot)$, at the value of 1. By explicitly setting $\xi(t) = \frac{\beta}{N} \sum_{n=0}^{N-1} t^n$, [22] obtained a tight upper bound β for γ^* .

Theorem 3 ([22, Th. 4]). *Suppose that the maps $\mathcal{M}_n$ satisfy the consistency property. Then, it holds that*

$$\mathbb{P}^N\{V(\hat{x}, \hat{y}) > g(s_N^*)\} \leq \beta,$$

where s_N^* is the complexity of δ^N and

$$g(k) = \begin{cases} 1 - t(k), & \text{if } k = 0, 1, \dots, N-1, \\ 1, & \text{if } k = N, \end{cases} \quad (2.6)$$

with $t(k)$ being the unique solution in the interval $(0, 1)$ of

$$\frac{\beta}{N} \sum_{n=k}^{N-1} \binom{n}{k} t^{n-k} - \binom{N}{k} t^{N-k} = 0.$$

It is worth noting that $\beta > \gamma^*$, but the gap is negligible. The powerfulness of Theorems 2 and 3 lies in their removal of the non-degeneracy assumption. Instead, only the consistency property is required here, which is generally milder and holds for a lot of (non-convex) problems. A direct application is non-convex scenario optimization, see [22, Sec. 3]. In cases where the consistency property is violated, the interested readers are referred to a recent paper [49] for a post-design verification methodology.

Lastly, we briefly introduce an earlier, complementary result in the context of non-convex scenario optimization.

Theorem 4 ([21, Th. 1]). *Given a multi-sample δ^N , suppose the map $\mathcal{M}_N$ satisfies $\mathcal{M}_N(\delta^N) \in \mathcal{X}_{\delta_i}$ for any $i \in \{1, \dots, N\}$. Then, it holds that*

$$\mathbb{P}^N\{V(x_N^*) < \epsilon\} \geq 1 - \beta,$$

where function $\epsilon : \{0, 1, \dots, N\} \rightarrow [0, 1]$ satisfies $\sum_{k=0}^{N-1} \binom{N}{k} (1 - \epsilon(k))^{N-k} = \beta$ with $\epsilon(N) = 1$.

Remark 1. *It is worth noting that the relaxed bound in Theorem 4 is valid under significantly milder assumptions compared to Theorem 2. Consequently, users can leverage either of these two bounds depending on the specific requirements and conditions of the problem at hand.*

2.2 Minimax Problems

Let $\mathcal{X} \subseteq \mathbb{R}^n$ and $\mathcal{Y} \subseteq \mathbb{R}^m$, and let $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ be a continuously differentiable payoff function. Minimax problems model a two-player zero-sum game where one player aims to minimize a payoff function while the other aims to maximize it. Namely,

$$\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y), \quad (2.7)$$

where min-player selects a strategy $x \in \mathcal{X}$ to minimize the payoff, while the max-player selects a strategy $y \in \mathcal{Y}$ to maximize it. This structure underpins a vast array of important applications, from optimal control [5] to machine learning [6]. In recent years, minimax problems get a lot of attention, largely due to the broad applications of generative adversarial networks (GANs) [9].

In this section, we introduce some basic equilibrium concepts and algorithms related to minimax problems.

2.2.1 Definitions of Equilibrium Concepts

In this subsection, we begin by revisiting the convex–concave case, where the payoff f is convex in x and concave in y , and the constraint sets $\mathcal{X}$ and $\mathcal{Y}$ are convex. In this setting, the minimax problem (2.7) is well studied [7, 8], with NE serving as a central concept.

Definition 4. A collective strategy (x^*, y^*) is an NE of problem (2.7) if, for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$:

$$f(x^*, y) \leq f(x^*, y^*) \leq f(x, y^*).$$

An NE is equivalent to a global saddle point, satisfying the condition that no player has incentive to unilaterally deviate. The subsequent lemma establishes sufficient conditions for the existence of an NE.

Lemma 1 ([7]). Suppose $\mathcal{X}$ and $\mathcal{Y}$ are convex, compact, and nonempty, and suppose the payoff f is convex–concave, i.e., f is convex in x and concave in y . Then, the minimax problem (2.7) admits an NE on $\mathcal{X} \times \mathcal{Y}$. Furthermore, the minimax equality holds

$$\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y) = \max_{y \in \mathcal{Y}} \min_{x \in \mathcal{X}} f(x, y).$$

In the convex–concave case, the minimax problem (2.7) can be framed as a *simultaneous game* where the order of moves does not matter.

Recent advances in machine learning have spurred growing interest in non-convex minimax problems. Prominent examples include GANs and adversarial training, which can often be formulated as non-convex minimax problems. Unfortunately, finding a global optimal solution efficiently is generally intractable, as non-convex optimization, which is already NP-hard, is a special case of non-convex minimax optimization. Consequently, researchers typically resort to seeking alternative points. However, due to the complex interaction between the min- and max- players, properly defining stationarity for NC–NC minimax problems remains a fundamental challenge, as pointed out by [50].

Among various equilibrium concepts, a critical one is ε -stationary point, which provides necessary conditions for a local NE. Following [51], we first define it as follows.

Definition 5. A collective strategy $(\hat{x}, \hat{y}) \in \mathcal{X} \times \mathcal{Y}$ is an ε -stationary point of (2.7), if

$$\begin{aligned} \text{dist}(0_n, \nabla_x f(\hat{x}, \hat{y}) + \mathcal{N}_{\mathcal{X}}(\hat{x})) &\leq \varepsilon, \\ \text{dist}(0_m, -\nabla_y f(\hat{x}, \hat{y}) + \mathcal{N}_{\mathcal{Y}}(\hat{y})) &\leq \varepsilon, \end{aligned}$$

where $\mathcal{N}_{\mathcal{S}}$ is the normal cone operator associated with a set $\mathcal{S} \subseteq \mathbb{R}^d$, and $\text{dist}(z, \mathcal{S}) := \inf_{v \in \mathcal{S}} \|z - v\|$ denotes the distance from $z \in \mathbb{R}^d$ to $\mathcal{S}$. The partial derivatives of f with respect to x and y are $\nabla_x f = \frac{\partial f}{\partial x}$ and $\nabla_y f = \frac{\partial f}{\partial y}$, respectively.

Remark 2. In this thesis, we only consider the Mordukhovich (limiting) normal cone, denoted by $\mathcal{N}_{\mathcal{X}}(x)$, which is defined as:

$$\mathcal{N}_{\mathcal{X}}(x) := \limsup_{x' \xrightarrow{\mathcal{X}} x} \widehat{\mathcal{N}}_{\mathcal{X}}(x'),$$

where $\widehat{\mathcal{N}}_{\mathcal{X}}(x')$ represents the Fréchet (regular) normal cone, and $\limsup$ denotes the Painlevé-Kuratowski outer limit (see, e.g., [52]). Unless otherwise specified, the term “normal cone” and the notation $\mathcal{N}_{\mathcal{X}}(x)$ refer strictly to the Mordukhovich normal cone.

It is shown in [53] that indeed GAN zero-sum games may have no Nash equilibria. Interestingly, the GANs framework implements a sequential two-player game: it first optimizes the discriminator D while keeping the generator G fixed, and then updates G using the optimized D . More generally, adversarial training describes the minimax problem (2.7) in the frame of a *sequential game*. These cases lead to the equilibrium concept known as *global minimax point*, which captures global optimality and is widely employed in machine

learning [54, 55, 56]. Intuitively, the min-player acts as the leader, first determining their optimal strategy, i.e., finding x^* such that $f(x^*, y^*) = \max_{y' \in \mathcal{Y}^N} f(x^*, y') \leq \max_{y' \in \mathcal{Y}^N} f(x, y')$, while the max-player then acts as the follower, choosing a best response to the min-player's choice, i.e., finding y^* such that $f(x^*, y) \leq f(x^*, y^*)$.

Following [54], we begin by defining the concept of global minimax point. Notice that it is enough to focus on a global minimax point of (2.7), since any global maximin point corresponds to its global minimax point (where the max-player is the leader and the min-player is the follower) by considering the negative payoff, $-f(x, y)$.

Definition 6. A collective strategy $(x^*, y^*) \in \mathcal{X} \times \mathcal{Y}$ is a global minimax point of (2.7), if for all (x, y) :

$$f(x^*, y) \leq f(x^*, y^*) \leq \max_{y' \in \mathcal{Y}} f(x, y').$$

In game-theoretic terms, this is also known as a *Stackelberg equilibrium*, where x is the leader while y is the follower. While there are certainly many other equilibrium concepts, they fall outside the scope of this thesis. Readers are kindly referred to [55] for a comprehensive overview.

2.2.2 Algorithms to Solve NC-NC Minimax Problems

Recent years have witnessed an explosion of literature on solving non-convex minimax problems. Given the vastness of this domain, this subsection focuses on two prominent lines of research: algorithms operating under one-sided dominance conditions, and those relying on the weak Minty variational inequality (MVI) framework.

One of the most straightforward algorithm is the gradient descent-ascent (GDA) method, which alternates simultaneously between a gradient descent step for x and a gradient ascent step for y . This is a natural generalization of gradient descent methods in minimization problems. However, GDA algorithm does not converge in the simplest bilinear function with $f(x, y) = xy$ (see, e.g., Fig. 2.2 and [55, Ex. 2.7]), let alone for NC-NC cases. Consequently, analyzing the convergence of GDA under specialized assumptions has become a mathematically tractable and highly active area of research within the machine learning community.

One-sided dominance condition. Recent literature has witnessed a strong interest in the dominance conditions, which include convexity, Polyak–Łojasiewicz (PŁ) condition, and the more general Kurdyka–Łojasiewicz (KŁ)

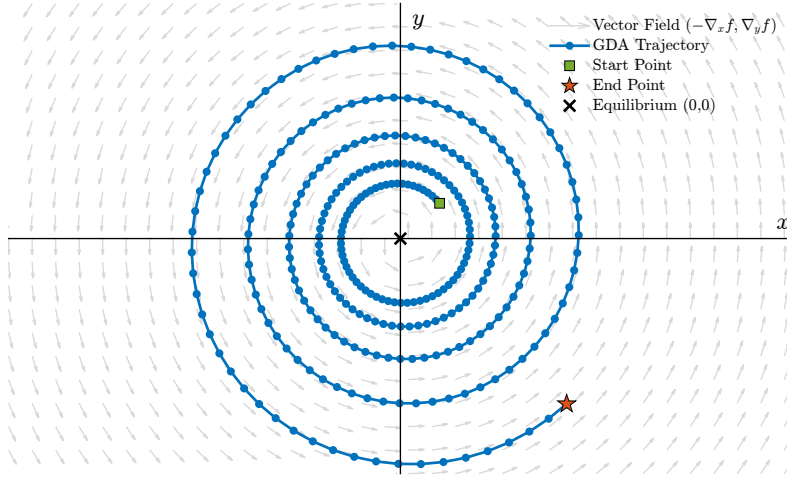


Figure 2.2: Divergence of GDA on payoff function $f(x, y) = xy$.

property on the primal (or dual) side. Given any $x \in \mathcal{X}$, assume the supremum of $f(x, \cdot)$ over $\mathcal{Y}$ is finite and attained. Furthermore, the payoff f is said to satisfy the dual side KL condition with parameter θ if, there exists $\mu > 0$ and $\theta \in (0, 1)$ such that

$$\left(\max_{y' \in \mathcal{Y}} f(x, y') - f(x, y) \right)^\theta \leq \frac{1}{\mu} \text{dist}(0_m, -\nabla_y f(x, y) + \mathcal{N}_{\mathcal{Y}}(y)).$$

In particular, this inequality degenerates into the PL condition when $\theta = 1/2$. From an algorithmic perspective, [51] proposed a doubly smoothed gradient descent-ascent method, which converges with an iteration complexity of $\mathcal{O}(\varepsilon^{-2 \max\{2\theta, 1\}})$. Later article [50] showed the same iteration complexity could be obtained by a smoothed proximal linear descent-ascent algorithm for the case of a nonsmooth composite primal objective. Moreover, it is proven that when $\theta \in [0, 1/2]$, this algorithm achieves the optimal iteration complexity of $\mathcal{O}(\varepsilon^{-2})$.

Weak Minty solutions. While the aforementioned one-side dominance condition asymmetrically exploits the geometric properties of one player's objective, another fundamental approach is to treat the minimax problem symmetrically by formulating it as a variational inequality (VI) problem. Let $F(z) := [\nabla_x f(x, y); -\nabla_y f(x, y)]$ for $z \in \mathcal{Z} := \mathcal{X} \times \mathcal{Y}$, the Stampacchia variational inequality problem is to find z^* such that

$$\langle F(z^*), z - z^* \rangle \geq 0, \forall z \in \mathcal{Z}. \quad (2.8)$$

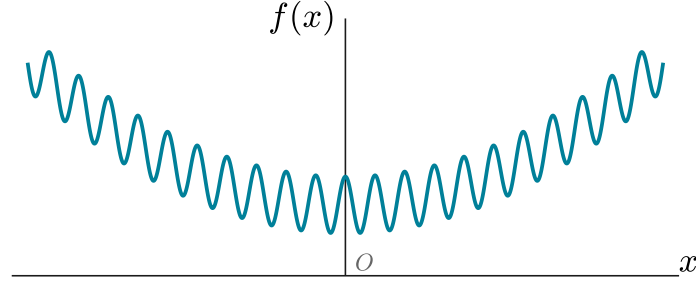


Figure 2.3: The landscape of a non-convex function satisfying ρ -comotonicity.

However, there is no guarantee that (2.8) admits least one solution in NC–NC minimax problems, since F is not monotone. A viable approach is to relax the nonnegative lower bound to allow for negative values, while replacing $F(z^*)$ with $F(z)$ in (2.8) for every $z \in \mathcal{Z}$. This leads to the weak Minty condition:

$$\langle F(z), z - z^* \rangle \geq -\rho \|F(z)\|^2, \quad \forall z \in \mathcal{Z}.$$

Regarding algorithms, [57] proposed a generalized extragradient algorithm for the case when $\rho \in [0, \frac{1}{4L})$ with iteration complexity of $\mathcal{O}(\varepsilon^{-2})$. Here, L is the Lipschitz constant of F . [58] later on proposed an adaptive extragradient algorithm extending the range of ρ to $[0, \frac{1}{L})$.

An intuitive way of understanding the weak MVI is to consider the unconstrained case* where $\mathcal{X} = \mathbb{R}^n$ and $\mathcal{Y} = \mathbb{R}^m$. The stationary solution z^* satisfies $F(z^*) = 0$, which implies the ρ -comonotonicity condition

$$\langle F(z) - F(z^*), z - z^* \rangle \geq -\rho \|F(z) - F(z^*)\|^2, \quad \forall z \in \mathbb{R}^{n+m}$$

is sufficient to find an MVI solution. Here, by allowing the parameter ρ to be strictly positive (yielding a negative lower bound), the operator F is no longer required to be monotone, which corresponds to a class of non-convex landscapes. Recall that a continuously differentiable function is convex (resp. strongly convex) if and only if its gradient is monotone (resp. strongly monotone). Therefore, such condition corresponds to a class of non-convexity. As the generic optimization problem in scalar case illustrated in Fig. 2.3, the landscape resembles "bent spring": while the global structure mimics a strongly convex function, the presence of numerous local minima renders it fundamentally non-convex.

Note that for constrained problems, the formulation requires adjusting the term $\|F(z)\|^2$ using the gradient mapping/natural residual, as the optimal operator $F(z^)$ may not be zero on the boundary.

Chapter 3

Robustness via the Scenario Approach

This chapter establishes rigorous and tight probabilistic guarantees for ε -stationary points over convex constraint sets. By demonstrating that the stationary residual is non-increasing as the number of scenarios increases, we show that ε -stationarity of these points persists as more appropriate scenarios are incorporated. Furthermore, we characterize the robustness of global minimax points under mild conditions. Shifting focus to nonconvex constraint sets, we identify fundamental challenges in obtaining tight theoretical bounds within the scenario approach; specifically, we show that the consistency property is not generally satisfied for global minimax points. To address this, we establish a relaxed, yet rigorously valid, probabilistic bound for global minimax points in the fully nonconvex setting. These results provide a theoretical foundation for applying the scenario approach to complex minimax problems where convexity–concavity assumptions are violated.

3.1 Problem Formulation

Let us consider a generic minimax problem subject to uncertain constraint sets:

$$\min_{x \in \mathcal{X}_\delta} \max_{y \in \mathcal{Y}_\delta} f(x, y), \quad (3.1)$$

where $\mathcal{X}_\delta \subseteq \mathbb{R}^n$ and $\mathcal{Y}_\delta \subseteq \mathbb{R}^m$ denote general uncertain constraint sets for the min-player and the max-player, respectively. Here, $\delta \in \Delta$ is an uncertain parameter governed by an arbitrary distribution, consistent with the framework introduced in Section 2.1.

In the scenario approach, the uncertainty in (3.1) is addressed by drawing a multi-sample of N i.i.d. scenarios, denoted by $\boldsymbol{\delta}^N = (\delta_1, \dots, \delta_N)$. By enforcing these N deterministic constraints simultaneously, we obtain the following deterministic scenario minimax problem:

$$\min_{x \in \mathcal{X}^N} \max_{y \in \mathcal{Y}^N} f(x, y), \quad (3.2)$$

where the feasible sets are defined as the intersection of all sampled constraints: $\mathcal{X}^N := \bigcap_{i=1}^N \mathcal{X}_{\delta_i}$ and $\mathcal{Y}^N := \bigcap_{i=1}^N \mathcal{Y}_{\delta_i}$. Throughout this chapter, we assume that $\mathcal{X}^N \times \mathcal{Y}^N$ is nonempty for any $N \in \mathbb{N}$ to ensure well-posedness.

To establish a theoretical baseline and contextualize our main results, we first introduce several standard assumptions and revisit a fundamental robustness result in the fully convex case from the existing literature.

Assumption 1 (Strategy convexity). $\mathcal{X}_{\delta}$ and $\mathcal{Y}_{\delta}$ are convex, compact, and nonempty for all $\delta \in \Delta$.

We further make the following non-degeneracy assumption. This is considered a mild assumption in convex settings, a point discussed comprehensively by [43, Sec. 8] and [22, Sec. 1.3].

Assumption 2 (Non-degeneracy). For any multi-sample $\boldsymbol{\delta}^N$ with $N \in \mathbb{N}$, with probability 1 over the random draws $\delta_1, \dots, \delta_N$, there exists a unique support list.

Given Lemma 1 and a tie-break rule ensuring uniqueness, e.g., by selecting the NE with minimum ℓ_2 -norm, a direct application of Theorem 1 yields a robustness guarantee for the NE.

Proposition 2. Let Assumptions 1 and 2 hold, suppose the payoff f is convex-concave, and let $h(i)$ be any $[0, 1]$ -valued function for $i = 0, 1, \dots, m + n$. Given a multi-sample $\boldsymbol{\delta}^N$, the NE (x^*, y^*) to the scenario minimax problem (3.2) satisfies

$$\mathbb{P}^N\{V(x^*, y^*) > h(s_N^*)\} \leq \gamma^*,$$

where s_N^* is the complexity of the multi-sample $\boldsymbol{\delta}^N$, and

$$\gamma^* = \inf_{\xi(\cdot) \in \mathbf{P}_N} \xi(1) \quad (3.3)$$

subject to

$$\frac{1}{k!} \frac{d^k}{dt^k} \xi(t) \geq \binom{N}{k} t^{N-k} \cdot \mathbb{1}\{t \in [0, 1 - h(k)]\}$$

for all $t \in [0, 1]$ and all $k = 0, 1, \dots, N$, where $\mathbb{1}\{\cdot\}$ denotes the indicator function, which equals 1 if the condition inside is true, and 0 otherwise; $\mathbf{P}_N$ denotes the class of polynomials of degree N .

Proposition 2 provides an intuitive insight: for convex–concave minimax problems subject to uncertain convex constraints, by applying suitable algorithms to the scenario minimax problem based on N scenarios, we can always find an NE with the minimum ℓ_2 norm. Furthermore, this NE is guaranteed to be robust against all uncertain constraints with high probability.

After revisiting the convex–concave case, we now turn to the central contribution of this thesis: establishing robustness results for NC–NC minimax problems. Notably, non-degeneracy in Assumption 2 is generally difficult to verify, even in convex optimization, and this challenge is amplified in the context of games [11]. Furthermore, the existence of an NE is no longer guaranteed once convexity is violated.

Accordingly, we state our problem as follows.

Problem 1. *For general NC–NC minimax problems, characterize the robustness of key equilibria.*

The remainder of this chapter investigates two distinct equilibrium concepts. Specifically, in Section 3.2, we leverage the foundational result of Theorem 2 to establish a tight probabilistic robustness guarantee for an ε -stationary point under convex constraint sets. Subsequently, in Section 3.3, we demonstrate that Theorem 2 is fundamentally inapplicable to fully non-convex settings in general, prompting us to derive an alternative robustness bound for a global minimax point. The quantitative relation between two probabilistic robustness bounds is also included.

3.2 NC–NC Minimax Problems with Convex Constraint Sets

This section investigates the probabilistic robustness guarantees for ε -stationary points in NC–NC minimax problems with convex constraint sets.

Quantifying the robustness of ε -stationary points is mathematically challenging in NC–NC case, where the complex geometry and potential for degeneracy make direct analysis intractable. Our core strategy is to first prove the existence of an ε -stationary point under Assumption 1, and then establish its consistency property introduced in Theorem 2 and in that way characterize

the stability of ε -stationary points against sample changes. Note that the result holds even without a non-degeneracy assumption.

Theorem 5. *Let Assumption 1 hold. For a given multi-sample δ^N , there exists an ε -stationary point $(\hat{x}, \hat{y})$ of (3.2) for some $\varepsilon > 0$ such that*

$$\mathbb{P}^N\{V(\hat{x}, \hat{y}) > h(s_N^*)\} \leq \gamma^*,$$

where $h(\cdot)$ is any $[0, 1]$ -valued function, s_N^* is the complexity of the multi-sample δ^N , and γ^* is defined in (3.3).

Proof. The ε -stationary point's invariance to permutation of the multi-sample δ^N means that condition 1 of the consistency property is naturally met.

To make the dependence on the sample set explicit for a fixed δ^N , define the stationary residual

$$r^N(x, y) := \max \left\{ \text{dist}(\mathbf{0}_n, \nabla_x f(x, y) + \mathcal{N}_{\mathcal{X}^N}(x)), \right. \\ \left. \text{dist}(\mathbf{0}_m, -\nabla_y f(x, y) + \mathcal{N}_{\mathcal{Y}^N}(y)) \right\}.$$

The continuous differentiability of f implies the continuity of both $\nabla_x f$ and $\nabla_y f$. By [52, Prop. 6.6], relative to $\mathcal{X}^N$, the set-valued mapping $x \mapsto \mathcal{N}_{\mathcal{X}^N}(x)$ is outer semicontinuous with closed values. Therefore, [52, Prop. 5.11] guarantees that for fixed y , the map

$$x \mapsto \text{dist}(\mathbf{0}_n, \nabla_x f(x, y) + \mathcal{N}_{\mathcal{X}^N}(x))$$

is lower semicontinuous (lsc) in x , and similarly for the y -component. Consequently, since $r^N(x, y)$ is the maximum of two lsc functions, it follows directly that $r^N(x, y)$ is lsc relative to (x, y) . Under Assumption 1, [52, Th. 1.9] guarantees that there exists

$$(\hat{x}_N, \hat{y}_N) \in \arg \min_{(x, y) \in \mathcal{X}^N \times \mathcal{Y}^N} r^N(x, y)$$

such that $r^N(\hat{x}_N, \hat{y}_N) = \varepsilon$.

Now consider augmenting δ^N by an additional multi-sample δ^K . Observe that both $\mathcal{X}^{N+K}$ and $\mathcal{X}^N$ are convex, and the subset relation $\mathcal{X}^{N+K} \subseteq \mathcal{X}^N$ leads to

$$\mathcal{N}_{\mathcal{X}^N}(x) \subseteq \mathcal{N}_{\mathcal{X}^{N+K}}(x), \quad \forall x \in \mathcal{X}^N, \quad (3.4)$$

and similarly for the y -component. Hence, for every (x, y) in the common

domain $\mathcal{X}^N \times \mathcal{Y}^N$, we have

$$\text{dist}(0_n, \nabla_x f(x, y) + \mathcal{N}_{\mathcal{X}^{N+K}}(x)) \leq \text{dist}(0_m, \nabla_x f(x, y) + \mathcal{N}_{\mathcal{X}^N}(x)),$$

and an analogous inequality holds for the y -component. It follows that for all (x, y) ,

$$r^{N+K}(x, y) \leq r^N(x, y).$$

Take the pair

$$(\hat{x}_{N+K}, \hat{y}_{N+K}) \in \arg \min_{(x, y) \in \mathcal{X}^{N+K} \times \mathcal{Y}^{N+K}} r^{N+K}(x, y)$$

as a minimizer for the augmented multi-sample δ^{N+K} . If for all $k \in \{1, \dots, K\}$ it holds that

$$(\hat{x}_N, \hat{y}_N) \in \mathcal{X}^{N+k} \times \mathcal{Y}^{N+k},$$

where $\mathcal{X}^{N+k} = \cap_{i=1}^{N+k} \mathcal{X}_{\delta_i}$ and $\mathcal{Y}^{N+k} = \cap_{i=1}^{N+k} \mathcal{Y}_{\delta_i}$, then in particular

$$(\hat{x}_N, \hat{y}_N) \in \mathcal{X}^{N+K} \times \mathcal{Y}^{N+K}$$

and therefore

$$r^{N+K}(\hat{x}_{N+K}, \hat{y}_{N+K}) \leq r^{N+K}(\hat{x}_N, \hat{y}_N) \leq r^N(\hat{x}_N, \hat{y}_N) = \varepsilon.$$

This shows that the ε -stationary point remains valid after adding new appropriate samples; in other words, the scenarios satisfy confirmation under appropriateness, the condition 2 of the consistency property.

Conversely, if for some $k \in \{1, \dots, K\}$ the point $(\hat{x}_N, \hat{y}_N)$ fails to belong to $\mathcal{X}^{N+k} \times \mathcal{Y}^{N+k}$, then in particular

$$(\hat{x}_N, \hat{y}_N) \notin \mathcal{X}^{N+K} \times \mathcal{Y}^{N+K}.$$

Recall that

$$(\hat{x}_{N+K}, \hat{y}_{N+K}) \in \mathcal{X}^{N+K} \times \mathcal{Y}^{N+K}.$$

Thus, we conclude that

$$(\hat{x}_N, \hat{y}_N) \neq (\hat{x}_{N+K}, \hat{y}_{N+K}),$$

thereby verifying the responsiveness to inappropriateness of scenarios.

Combining the above observations with Theorem 2 completes the proof. $\square$

Theorem 5 establishes a generalization guarantee of an ε -stationary point to problem (3.2). Given a solution $(\hat{x}_N, \hat{y}_N)$ obtained via a suitable algorithm, such as the one proposed in the next subsection, Theorem 5 provides a probabilistic upper bound on its generalization error. Notably, while our proof starts from the existence of an ε -stationary point, the resulting robustness bound is algorithm-independent and valid for any point meeting this condition. For instance, one could first determine a desired small ε and employ a suitable algorithm, such as the one proposed in [51], to reach this threshold. Consequently, treating this achieved residual as ε , the remainder of the theoretical guarantees in Theorem 5 follows naturally.

From a practical perspective, one typically fixes a high confidence level $1 - \beta$ (i.e., very close to 1) and, in return, desires a deterministic upper bound $h(s_N^*)$ for the probability of violation $V(\hat{x}_N, \hat{y}_N)$. Since the function $h(\cdot)$ as in Theorem 2 and Theorem 5 is indefinite for a fixed γ^* , there may exist infinitely many solutions $\varepsilon(k)$ to (3.3). A practical corollary resolves this issue by directly applying Theorem 3, as follows.

Corollary 1. *Under Assumption 1, for a given $\beta \in (0, 1)$ and a multi-sample δ^N , the ε -stationary point $(\hat{x}, \hat{y})$ of (3.2) satisfies*

$$\mathbb{P}^N\{V(\hat{x}, \hat{y}) > g(s_N^*)\} \leq \beta,$$

where s_N^* is the complexity of δ^N and

$$g(k) = \begin{cases} 1 - t(k), & \text{if } k = 0, 1, \dots, N-1, \\ 1, & \text{if } k = N, \end{cases} \quad (3.5)$$

with $t(k)$ being the unique solution in the interval $(0, 1)$ of

$$\frac{\beta}{N} \sum_{n=k}^{N-1} \binom{n}{k} t^{n-k} - \binom{N}{k} t^{N-k} = 0.$$

Equivalently, the result in Corollary 1 can be stated in terms of the complementary probability:

$$\mathbb{P}^N\{V(\hat{x}, \hat{y}) \leq g(s_N^*)\} \geq 1 - \beta.$$

This probabilistic bound gives a powerful robustness guarantee for an obtained ε -stationary point $(\hat{x}, \hat{y})$ from a multi-sample δ^N . By first selecting an arbitrarily high confidence level $1 - \beta^*$ and then computing the complexity s_N^*

*Numerical results suggest that β could be chosen as small as possible since $g(k)$ exhibits

of δ^N , one can uniquely determine the upper bound $g(s_N^*)$ using (3.5). That is, we have at least $1 - \beta$ confidence to ensure that $V(\hat{x}, \hat{y})$ will not exceed this computed and unique $g(s_N^*)$. Note that the probability of violation $V(\hat{x}, \hat{y})$ serves as a measure of robustness: the smaller its value, the more robust the ε -stationary point $(\hat{x}, \hat{y})$ is. Therefore, Corollary 1 formalizes this probabilistic robustness guarantee.

3.3 NC–NC Minimax Problems with Non-convex Constraint Sets

Let us consider the minimax problem (3.2) with NC–NC payoffs and non-convex constraint sets. In Section 3.2, we investigate the robustness of an ε -stationary point by playing a simultaneous game. However, in this simultaneous setting, it is not straightforward to extend the result of Theorem 5 to minimax problems with non-convex constraint sets. This is because the monotonicity of the stationary residual r^N with respect to the number of scenarios is difficult to maintain when the convexity assumption on the constraint sets (Assumption 1) is violated.

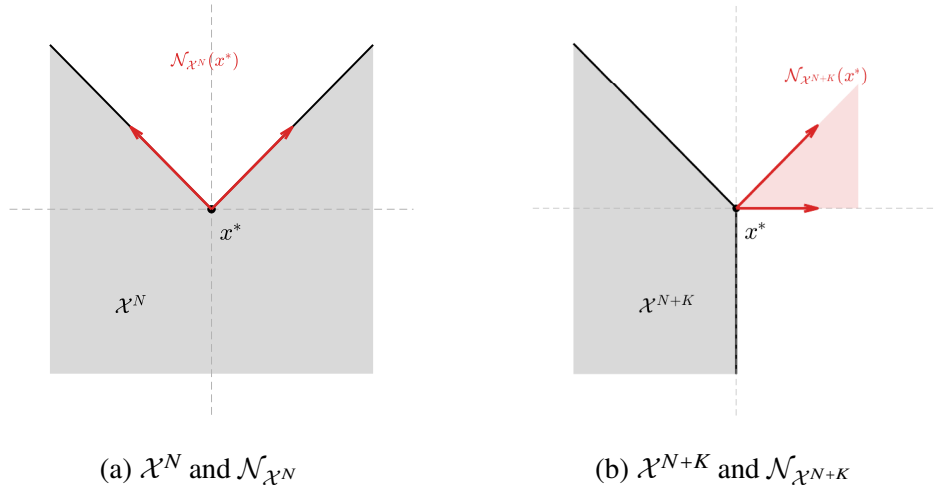


Figure 3.1: Geometric illustration of non-monotonicity of Mordukhovich normal cones (see, e.g., [52, Def. 6.3]) in non-convex sets. In the left and right panels, the red lines and the pink area represent $\mathcal{N}_{\mathcal{X}^N}(x^*)$ and $\mathcal{N}_{\mathcal{X}^{N+K}}(x^*)$, respectively. Clearly, $\mathcal{X}^{N+K} \subseteq \mathcal{X}^N$ does not imply $\mathcal{N}_{\mathcal{X}^N}(x^*) \subseteq \mathcal{N}_{\mathcal{X}^{N+K}}(x^*)$.

a modest dependence on the value of β , see [22, Sec. 2.1]. In other words, the values of $g(k)$ are almost the same for $\beta = 10^{-4}$ and $\beta = 10^{-8}$.

To see this easily, we notice that the set-theoretic inclusion $\mathcal{X}^{N+K} \subseteq \mathcal{X}^N$ remains valid, the corresponding monotonicity of normal cones, as shown in (3.4), however, loses its structural stability in the non-convex setting. See Figure 3.1 for a simple example. Without the support of convexity, the local geometry of the intersection set $\mathcal{X}^N$ can become arbitrarily complex, causing the stationary residual to fluctuate rather than remain non-increasing. This lack of regularity is precisely why the standard scenario framework is difficult to provide tight guarantees for NC–NC minimax problems with non-convex constraint sets.

Interestingly, the GANs framework implements a sequential two-player game: it first optimizes the discriminator D while keeping the generator G fixed, and then updates G using the optimized D . More generally, adversarial training describes the minimax problem (3.2) in the frame of a *sequential game*. These cases lead to the equilibrium concept known as *global minimax point*, which captures global optimality and is widely employed in machine learning [54, 55, 56]. Intuitively, the min-player acts as the leader, first determining their optimal strategy, while the max-player then acts as the follower, choosing a best response to the min-player's choice. For a game-theoretic perspective, this setup corresponds to a Stackelberg game.

The next assumption removes all convexity requirements, including the constraint sets originally assumed to be convex in Assumption 1.

Assumption 3 (Strategy non-convexity). $\mathcal{X}_\delta$ and $\mathcal{Y}_\delta$ are compact and nonempty for all $\delta \in \Delta$.

The following lemma, proven by the extreme value theorem, guarantees the existence of a global minimax point.

Lemma 2 ([54]). Under Assumption 3, there exists at least one global minimax point (x^*, y^*) of the scenario minimax problem (3.2).

The following proposition states the difficulty of establishing a tight bound for a global minimax point by utilizing the latest result [22].

Remark 3. It is important to note that in the absence of further structural assumptions beyond Assumption 3, the consistency property for a global minimax point does not hold in general (Theorem 2).

To see this, let us define the envelope functions

$$\phi_N(x) := \max_{y' \in \mathcal{Y}^N} f(x, y'), \quad \phi_{N+K}(x) := \max_{y' \in \mathcal{Y}^{N+K}} f(x, y').$$

By Definition 6, given a multi-sample δ^N , the global minimax point (x_N^*, y_N^*) satisfies

$$\phi_N(x_N^*) \leq \phi_N(x), \quad \forall x \in \mathcal{X}^N.$$

Augmenting δ^N by an additional multi-sample δ^K , and assuming $(x_N^*, y_N^*) \in \mathcal{X}^{N+K} \times \mathcal{Y}^{N+K}$, we have

$$\phi_{N+K}(x_N^*) = \max_{y' \in \mathcal{Y}^{N+K}} f(x_N^*, y') = \phi_N(x_N^*),$$

where the last equality holds because y_N^* is the maximizer over the larger feasible set $\mathcal{Y}^N$, and it remains feasible in the smaller set $\mathcal{Y}^{N+K}$.

The consistency property holds if, at least, the original x_N^* satisfies

$$\phi_{N+K}(x_N^*) \leq \phi_{N+K}(x), \quad \forall x \in \mathcal{X}^{N+K}.$$

However, the two available conditions:

1. $\phi_{N+K}(x_N^*) = \phi_N(x_N^*) \leq \phi_N(x), \quad \forall x \in \mathcal{X}^{N+K};$
2. $\phi_{N+K}(x) \leq \phi_N(x), \quad \forall x \in \mathcal{X}^{N+K},$

cannot conclude $\phi_{N+K}(x_N^*) \leq \phi_{N+K}(x)$ in general. Namely, a global minimax point does not necessarily preserve its global optimality after adding new appropriate samples. Hence, the consistency property does not hold.

Given the lack of consistency for the global minimax point, we now turn our attention to its probabilistic robustness by leveraging an earlier foundational result [21], as stated in Theorem 3.

Theorem 6. *Let Assumption 3 hold. For a given multi-sample δ^N , the global minimax point (x^*, y^*) of the scenario minimax problem (3.2) satisfies*

$$\mathbb{P}^N\{V(x^*, y^*) > \epsilon(s_N^*)\} \leq \beta,$$

where $\epsilon(k) = 1 - \exp\left(\frac{1}{N-k} \ln\left(\frac{\beta}{N \binom{N}{k}}\right)\right)$ for $k = 0, 1, \dots, N-1$, with $\epsilon(N) = 1$.

Proof. Lemma 2, proven by the extreme value theorem, ensures the existence of a global minimax point, but not its uniqueness. To overcome this issue, we equip a suitable tie-breaking rule to $\mathcal{M}_N$, such as choosing a point (x^*, y^*) with minimum ℓ_2 -norm. Thus,

$$\mathcal{M}_N(\delta_1, \dots, \delta_N) = (x^*, y^*) \in \mathcal{X}^N \times \mathcal{Y}^N = \bigcap_{i=1}^N \mathcal{X}_{\delta_i} \times \bigcap_{i=1}^N \mathcal{Y}_{\delta_i},$$

where the last inequality is by notational convenience, which in turn implies $\mathcal{M}_N(\delta_1, \dots, \delta_N) \in \mathcal{X}_{\delta_i} \times \mathcal{Y}_{\delta_i}$ for all $i \in \{1, \dots, N\}$. Applying Theorem 3 yields $\sum_{k=0}^{N-1} \binom{N}{k} (1 - \epsilon(k))^{N-k} = \beta$, with $\epsilon(N) = 1$. As indicated in [21], setting each term in the summation equal to β/N leads to the exponential-logarithmic form of $\epsilon(k)$ by taking the logarithm on both sides. That is, we set

$$\binom{N}{k} (1 - \epsilon(k))^{N-k} = \frac{\beta}{N},$$

and taking the natural logarithm on both sides introduces to $(N - k) \ln(1 - \epsilon(k)) = \ln\left(\frac{\beta}{N \binom{N}{k}}\right)$. The proof is completed by exponentiating both sides of $\ln(1 - \epsilon(k)) = \frac{1}{N-k} \ln\left(\frac{\beta}{N \binom{N}{k}}\right)$. $\square$

While Theorem 5 focuses on a local ϵ -stationary point, Theorem 6 shows that a different high-probability generalization bound also holds for a global minimax point, even in fully non-convex settings. By calculating $\epsilon(s_N^*)$ —which is typically small in practice—one can assert, with probability at least $1 - \beta$, that the probability of violation $V(x^*, y^*)$ does not exceed $\epsilon(s_N^*)$. However, as a natural trade-off for achieving global optimality in fully non-convex problems, the generalization bound derived in Theorem 6 is inherently more conservative than that in Theorem 5. Although this fact is explicitly pointed out by [22], we give a brief proof here mathematically.

Proposition 3. *For a given $\beta \in (0, 1)$, it holds that for all $k = 0, 1, \dots, N - 2$,*

$$\epsilon(k) > g(k),$$

where $\epsilon(\cdot)$ is defined in Theorem 6, $g(\cdot)$ is defined in Theorem 5. Furthermore, $\epsilon(k) = g(k)$ for $k = N - 1$ and N .

Proof. Let function $c : \{0, 1, \dots, N - 1\} \rightarrow [0, 1]$ be

$$\begin{aligned} c(k) &:= 1 - \epsilon(k) \\ &= \exp\left(\frac{1}{N-k} \ln\left(\frac{\beta}{N \binom{N}{k}}\right)\right) \\ &= \exp\left(\ln\left(\left[\frac{\beta}{N \binom{N}{k}}\right]^{\frac{1}{N-k}}\right)\right) \\ &= \left(\frac{\beta}{N \binom{N}{k}}\right)^{\frac{1}{N-k}}. \end{aligned}$$

We therefore obtain $c(k)^{N-k} = \frac{\beta}{N \binom{N}{k}}$. Notice that in Theorem 5, we have

$$t(k)^{N-k} = \frac{\beta}{N \binom{N}{k}} \sum_{n=k}^{N-1} \binom{n}{k} t(k)^{n-k} = c(k)^{N-k} \sum_{n=k}^{N-1} \binom{n}{k} t(k)^{n-k}.$$

There are three cases:

1. $k = 0, 1, \dots, N-2$: it holds that $\sum_{n=k}^{N-1} \binom{n}{k} t(k)^{n-k} = \binom{k}{k} t(k)^0 + \sum_{n=k+1}^{N-1} \binom{n}{k} t(k)^{n-k} > 1$, which implies $t(k)^{N-k} > c(k)^{N-k}$. Since $N-k \geq 2$, we further have $t(k) > c(k)$ and $\epsilon(k) > g(k)$.
2. $k = N-1$: letting $k = N-1$ directly gives $\sum_{n=N-1}^{N-1} \binom{N-1}{N-1} t(k)^0 = 1$, which leads to $\epsilon(k) = g(k)$.
3. $k = N$: $\epsilon(k) = g(k)$ by definition.

This completes the proof. □

Finally, it is worth noting that without further assumptions, no general algorithms exist to locate such global minimax points. Given the inherent intractability of global non-convex optimization, computing global minimax points in general NC-NC settings is computationally prohibitive for universally applicable non-heuristic algorithms. Therefore, identifying tractable subclasses by exploiting specific structural or network architectures represents an important direction for future work.

Chapter 4

Numerical Experiments

This chapter presents two detailed numerical experiments to validate the theoretical results established in Chapter 3. Specifically, we illustrate Theorem 5 by introducing a specialized algorithm capable of finding an ε -stationary point for any user-prescribed tolerance ε , which can be arbitrarily small. Additionally, a zeroth-order heuristic algorithm for locating global minimax solutions is proposed, and applied to a concrete numerical case study.

4.1 Verification of Probabilistic Guarantees for ε -Stationary Points

In this section, we consider a concrete minimax problem with one-sided dominance condition. A first-order algorithm is employed to find an ε -stationary point with a desired value of ε . Finally, probabilistic validation of Theorem 5 is conducted under different value of confidence levels.

4.1.1 Problem Setup and Scenario Generation

Consider the NC–NC minimax problem with two-sided KL property [59]

$$\min_{x \in \mathcal{X}_\delta} \max_{y \in \mathcal{Y}_\delta} f(x, y) = x^2 + 3(\sin x)^2(\sin y)^2 - 4y^2 - 10(\sin y)^2,$$

where the uncertainty sets $\mathcal{X}_\delta = \mathcal{Y}_\delta = [0.2 - \delta, 2.2 + \delta]$ with δ uniformly distributed in $\Delta = [0.1, 0.5]$. It should be noted that such a uniform distribution is not necessary for the application of the scenario approach or for the results presented in Chapter 3; rather, it is considered only as a special numerical case for validating the theoretical results.

To see the geometry with respect to x , note that for any fixed y , the partial gradient $\nabla_x f(x, y) = 2x + 3 \sin^2(y) \sin(2x)$ is strictly positive over the feasible set $\mathcal{X}_\delta$ (since $x \geq 0.1$). Consequently, the function is strictly monotonically increasing with respect to x , and the unique constrained minimum is attained at the boundary $x^* = 0.2 - \delta$. Due to this strict monotonicity and bounded domain, the suboptimality gap is bounded by the generalized gradient distance (incorporating the normal cone $\mathcal{N}_{\mathcal{X}_\delta}(x)$), analogously satisfying the KL property on $\mathcal{X}_\delta$. A similar argument for the y -direction is omitted.

To construct the scenario-based counterpart of this uncertain problem, we generate a set of N i.i.d. samples of the uncertain parameter δ . Specifically, we draw samples $\{\delta_1, \delta_2, \dots, \delta_N\}$ from the uniform distribution on $\Delta = [0.1, 0.5]$. Each extracted sample δ_i yields a deterministic realization of the uncertainty sets, denoted as $\mathcal{X}_{\delta_i}$ and $\mathcal{Y}_{\delta_i}$. For instance, in one run, one may obtain $\delta_j = 0.3$, which leads to $\mathcal{X}_{\delta_j} = \mathcal{Y}_{\delta_j} = [-0.1, 2.5]$. By leveraging these generated scenarios, the original problem is transformed into a tractable optimization model based on finite empirical data:

$$\min_{x \in \mathcal{X}^N} \max_{y \in \mathcal{Y}^N} x^2 + 3(\sin x)^2(\sin y)^2 - 4y^2 - 10(\sin y)^2,$$

where $\mathcal{X}^N = \mathcal{X}_{\delta_1} \times \dots \times \mathcal{X}_{\delta_N}$ and $\mathcal{Y}^N = \mathcal{Y}_{\delta_1} \times \dots \times \mathcal{Y}_{\delta_N}$.

4.1.2 Computational Method: A First-Order Algorithm

In this section, we employ the doubly smoothed GDA algorithm in [51], that is,

$$\begin{aligned} x_{k+1} &= \text{Proj}_{\mathcal{X}^N} [x_k - \tau \nabla_x f(x_k, y_k) - \rho(x_k - \tilde{x}_k)], \\ \tilde{x}_{k+1} &= \tilde{x}_k - \rho(\tilde{x}_k - x_{k+1}), \\ y_{k+1} &= \text{Proj}_{\mathcal{Y}^N} [y_k + \tau \nabla_y f(x_{k+1}, y_k) - \rho(y_k - \tilde{y}_k)], \\ \tilde{y}_{k+1} &= \tilde{y}_k - \rho(\tilde{y}_k - y_{k+1}) \end{aligned}$$

to iteratively compute the ϵ -stationary point. Here, $\text{Proj}_{\mathcal{Z}} : \mathbb{R}^d \rightarrow \mathcal{Z}$ denotes the Euclidean projection operator onto the set $\mathcal{Z} \subseteq \mathbb{R}^d$, defined as $\text{Proj}_{\mathcal{Z}}(x) := \arg \min_{z \in \mathcal{Z}} \|z - x\|_2$. To better understand the rationale behind this iterative scheme, the updates can be interpreted as a regularized descent-ascent process. The projected gradient steps for x and y essentially denote the primal descent and dual ascent, respectively. Meanwhile, the auxiliary sequences $\{\tilde{x}_k\}$ and $\{\tilde{y}_k\}$ serve as proximal tracking mechanisms. These proximal steps are specifically introduced to counteract the inherent “non-convexity” and “non-

concavity” of the objective landscape, stabilizing the otherwise oscillatory minimax dynamics. Crucially, the deliberately shifted domain (incorporating the 0.2 and 2.2 offsets) ensures that the unconstrained saddle point is pushed toward or outside the boundary of the feasible region. This design forces the constraints to become active, preventing a trivially safe solution and allowing us to meaningfully evaluate the out-of-sample risk.

In the above updates, $\tau > 0$ denotes the primal-dual step size, and $\rho > 0$ represents the smoothing parameter that controls the strength of the double moreau envelope. The sequences $\{\tilde{x}_k\}$ and $\{\tilde{y}_k\}$ serve as auxiliary smoothing variables to stabilize the minimax optimization process. For the initialization, we set the auxiliary variables to coincide with the starting points, i.e., $\tilde{x}_0 = x_0$ and $\tilde{y}_0 = y_0$, which are randomly sampled from the feasible sets $\mathcal{X}^N$ and $\mathcal{Y}^N$.

In practice, the iterative process is terminated when the norm of the gradient mapping falls below the predefined residual/tolerance ϵ . Specifically, the algorithm stops when the condition

$$\max \left\{ \text{dist}(0_n, \nabla_x f(x_k, y_k) + \mathcal{N}_{\mathcal{X}^N}(x_k)), \right. \\ \left. \text{dist}(0_m, -\nabla_y f(x_k, y_k) + \mathcal{N}_{\mathcal{Y}^N}(y_k)) \right\} \leq \epsilon$$

is met, indicating that a valid ϵ -stationary point has been reached. In the numerical experiments, we always set the residual to $\epsilon = 10^{-2}$.

4.1.3 Probabilistic Validation and Results

In practical non-convex scenario optimization, computing the complexity s_N^* represents an extremely critical procedure, since *a priori* bound like the result in [38] is hard to obtain. In this thesis, we employ the greedy algorithm proposed in [21, Sec. 2]. While this algorithm is highly versatile and applicable to arbitrary (non-convex) optimization problems, its computational burden remains relatively heavy. To address this, several ad-hoc alternative algorithms have been developed to evaluate complexity, such as the one introduced in [11]. A deeper discussion of these alternatives is omitted here, as it falls outside the scope of this thesis.

We note the empirical and so average risks do not rely on the confidence level $1 - \beta$, while the theoretical upper bound $g(s_n^*)$ does. The experimental results in are shown in Fig. 4.1 on the probabilistic robustness of the obtained ϵ -stationary point (x_ϵ, y_ϵ) using the scenario approach. In the figure, the red, orange and purple lines represent the theoretical upper bound of the risk, denoted as $g(s_N^*)$ with $\beta = 10^{-1}, 10^{-2}, 10^{-4}$, respectively. The blue dashed

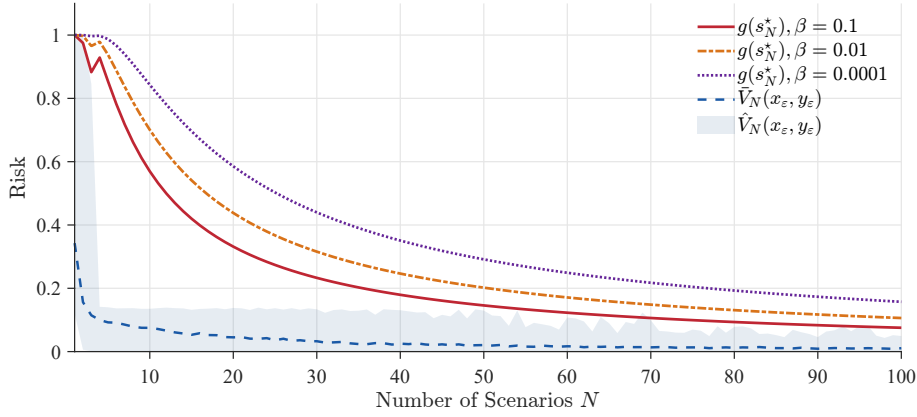


Figure 4.1: The risk depends on the number of scenarios for $\beta = 10^{-1}, 10^{-2}, 10^{-4}$. The figure compares the theoretical bound $g(s_N^*)$ (red solid line, orange dashed line, purple dashed line) with the range of all possible empirical risks $\hat{V}_N(x_\varepsilon, y_\varepsilon)$ (light grey shaded area) and the average empirical risk $\bar{V}_N(x_\varepsilon, y_\varepsilon)$ (blue dashed line) across different numbers of scenarios.

line represents the average empirical risk $\bar{V}_N(x_\varepsilon, y_\varepsilon)$, while the light blue shaded region illustrates the spread of the empirical risk $\hat{V}_N(x_\varepsilon, y_\varepsilon)$ across multiple $R = 200$ Monte Carlo repetitions.

As observed, while the theoretical bound $g(s_N^*)$ decays gradually with an increasing number of scenarios N , from 1 to 100, the empirical risk $\bar{V}_N(x_\varepsilon, y_\varepsilon)$ exhibits a sharp initial drop and consistently remains strictly bounded below the theoretical curve for $\beta = 10^{-2}, 10^{-4}$. When $\beta = 10^{-1}$, only once of violation is observed around $N = 69$: the true confidence level is $1 - 1/(200 \times 100) = 99.995\% \geq 1 - \beta = 90\%$. The results in Fig. 4.1 validate the probabilistic robustness of the obtained ε -stationary point $(x_\varepsilon, y_\varepsilon)$ using the scenario approach.

4.2 Verification of Probabilistic Guarantees for Global Minimax Points

In this subsection, we consider a bilinearly-coupled minimax problem. A zeroth-order heuristic algorithm is proposed to find a global minimax point. Lastly, we numerically verify the probabilistic guarantees of Theorem 6 by varying the confidence levels, and compare two different probabilistic robustness bounds.

4.2.1 Problem Setup and Scenario Generation

Consider an NC–NC minimax problem with bilinear interaction [60]

$$\begin{aligned} \min_{x \in \mathcal{X}_\delta} \max_{y \in \mathcal{Y}_\delta} f(x, y) = & (x + 1)(x + 3)(x - 1)(x - 3) \\ & + 10xy \\ & - (y + 1)(y + 3)(y - 1)(y - 3), \end{aligned} \quad (4.1)$$

where the uncertainty sets $\mathcal{X}_\delta = \mathcal{Y}_\delta = [-2\delta, -\delta] \cup [\delta, 2\delta]$ with δ uniformly distributed in $[1.0, 1.5]$. Notably, this payoff does not satisfy any established regularity conditions, such as the KL condition, the weak Minty condition, or others in the literature.

First, the objective $f(x, y)$ in (4.1) violates the KL property. The coupling of the min-player’s non-convex double-well potential and the max-player’s non-concave potential via the strong bilinear term $10xy$ yields a highly irregular landscape. This structure induces multiple spurious local minimax points and saddles. Near these regions, the respective gradients scale disproportionately or cancel out, preventing the existence of a uniform KL exponent to bound the gradient dynamics.

Second, the problem fails to satisfy the weak MVI condition. The strong coupling term $10xy$ severely distorts the gradient vector field, inducing prominent rotational dynamics. Consequently, rather than pointing toward the global optimum, gradient-based trajectories tend to swirl around or diverge from local saddles indefinitely. This active rotation away from the solution set violates the inner-product inequality fundamental to the weak MVI framework.

Finally, applying the standard scenario approach yields the following scenario minimax problem:

$$\min_{x \in \mathcal{X}^N} \max_{y \in \mathcal{Y}^N} (x + 1)(x + 3)(x - 1)(x - 3) + 10xy - (y + 1)(y + 3)(y - 1)(y - 3),$$

where $\mathcal{X}^N = \mathcal{X}_{\delta_1} \times \cdots \times \mathcal{X}_{\delta_N}$ and $\mathcal{Y}^N = \mathcal{Y}_{\delta_1} \times \cdots \times \mathcal{Y}_{\delta_N}$.

4.2.2 Computational Method: A Zeroth-Order Algorithm

According to Definition 6, locating a global minimax point requires solving the outer minimization globally over the entire feasible set $\mathcal{X}^N$. Namely, given the scenario minimax problem associated with a multi-sample $\boldsymbol{\delta}^N$, we compute the empirical worst-case (envelope) objective $\phi_N(x) = \max_{y \in \mathcal{Y}^N} f(x, y)$. The

Algorithm 1 Multi-Sample Zeroth-Order Global Minimax Search

Require: multi-sample δ^N , iterations T , inner size J , gradient samples M
Require: stepsizes η_x, η_y , smoothing μ_x, μ_y , variance schedule $\{\sigma_t\}$

- 1: Initialize $x_0 \in \mathcal{X}^N, y_0 \in \mathcal{Y}^N$
- 2: **for** $t = 0$ to $T - 1$ **do**
- 3: *// 1. Global Exploration*
- 4: Generate candidates $\mathcal{S}_t = \{y_t + \sigma_t \xi_j\}_{j=1}^J$ with $\xi_j \sim \text{Gaussian}(0, I)$
- 5: Select base point: $\tilde{y}_t \leftarrow \arg \max_{y \in \mathcal{S}_t} f(x_t, y)$
- 6: *// 2. ZO Ascent for y*
- 7: Estimate $\hat{g}_y = \frac{1}{M} \sum_{i=1}^M \frac{f(x_t, \tilde{y}_t + \mu_y v_i) - f(x_t, \tilde{y}_t - \mu_y v_i)}{2\mu_y} v_i$ using random $\{v_i\}$
- 8: Update $y_{t+1} \leftarrow \text{Proj}_{\mathcal{Y}^N}(\tilde{y}_t + \eta_y \hat{g}_y)$
- 9: *// 3. ZO Descent for x*
- 10: Estimate $\hat{g}_x = \frac{1}{M} \sum_{i=1}^M \frac{f(x_t + \mu_x u_i, y_{t+1}) - f(x_t - \mu_x u_i, y_{t+1})}{2\mu_x} u_i$ via random $\{u_i\}$
- 11: Update $x_{t+1} \leftarrow \text{Proj}_{\mathcal{X}^N}(x_t - \eta_x \hat{g}_x)$
- 12: **end for**
- 13: **return** (x_N^*, y_N^*) where $x_N^* = x_T$ and $y_N^* = \arg \max_{y \in \mathcal{Y}^N} f(x_T, y)$

min-player's strategy is selected by solving the global optimization problem:

$$x_N^* \in \arg \min_{x \in \mathcal{X}^N} \phi_N(x) = \arg \min_{x \in \mathcal{X}^N} \left(\max_{y \in \mathcal{Y}^N} f(x, y) \right),$$

while the max-player chooses the corresponding worst-case strategy:

$$y_N^* \in \arg \max_{y \in \mathcal{Y}^N} f(x_N^*, y).$$

With this global minimax point (x_N^*, y_N^*) defined, the primary computational challenge is effectively exploring the NC–NC landscape to avoid local optima. Note that while Berge's maximum theorem guarantees that the envelope function $\phi_N(x)$ is continuous, it is possibly nonsmooth. To address this, we propose the Multi-Sample Zeroth-Order Global Minimax Search (Algorithm 1).

The design of our algorithm is inspired by recent advances in *derivative-free optimization* (also known as zeroth-order optimization). This framework has proven highly effective in robust settings where exact gradients are unavailable, such as generating black-box adversarial attacks [61]. To address the non-differentiable issues induced by the nonsmooth maximum function $\phi_N(x)$, we employ the Gaussian smoothing technique. The theoretical foundation for such gradient-free approximation via Gaussian smoothing

was rigorously established by Nesterov and Spokoiny [62]. Building upon these foundations, our algorithm incorporates dynamic global exploration alongside alternating zeroth-order updates to aggressively escape spurious local stationary points in the non-convex landscape.

Instead of relying on multi-start heuristics, our method explicitly incorporates global exploration into each iteration. At iteration t , we first construct a perturbed candidate set $\mathcal{S}_t \subset \mathcal{Y}^N$ of size J to reduce the risk of being trapped in poor local optima in the inner maximization. The best candidate $\tilde{y}_t$ is used as the starting point for subsequent updates. We then estimate zeroth-order gradients for both variables using finite differences along M Gaussian directions. The updates for y and x are performed via projected zeroth-order ascent and descent methods, respectively. This combination of random exploration and zeroth-order updates provides a practical approach to approximating the global minimax solution (x_N^*, y_N^*) , as long as iterations T , inner size J and gradient samples M are large enough.

4.2.3 Probabilistic Validation and Results

With this minimax point (x_N^*, y_N^*) , we aim to ensure that $\mathbb{P}\{\delta \in \delta : (x_N^*, y_N^*) \notin \mathcal{X}_\delta \times \mathcal{Y}_\delta\} > \epsilon(s_N^*)$ holds with probability at most β , where the complexity s_N^* is evaluated using the greedy algorithm described in [21, Sec. 2]. To empirically validate this bound $\epsilon(s_N^*)$, a Monte Carlo simulation with $R = 100$ repetitions

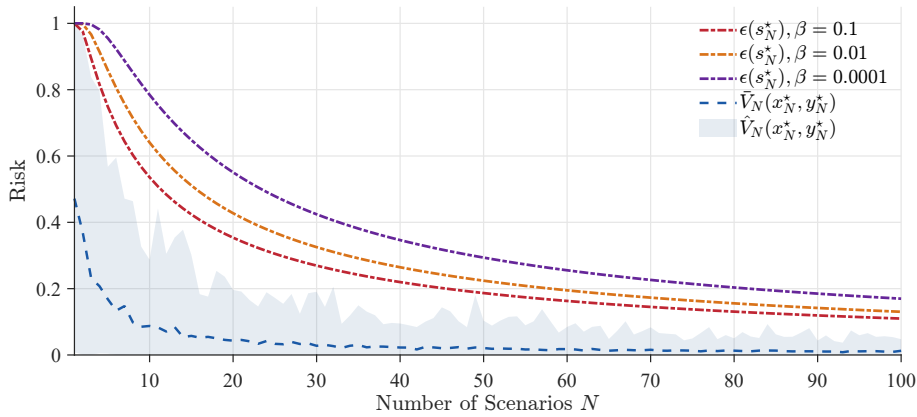


Figure 4.2: The risk depends on the number of scenarios for $\beta = 10^{-1}, 10^{-2}, 10^{-4}$. The figure compares the theoretical bound $\epsilon(s_N^*)$ (red, orange, and purple dashed line) with the range of all possible empirical risks $\hat{V}_N(x_N^*, y_N^*)$ (light grey shaded area) and the average empirical risk $\bar{V}_N(x_N^*, y_N^*)$ (blue dashed line) across different numbers of scenarios.

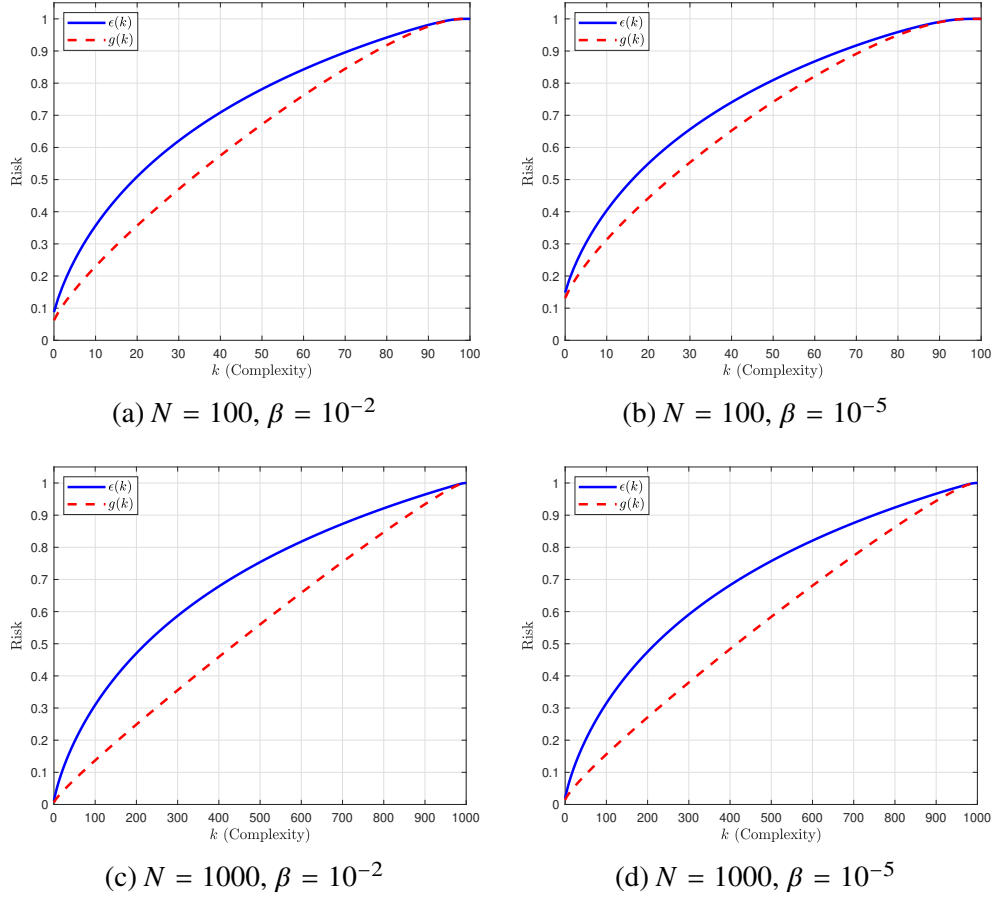


Figure 4.3: Comparison of the explicit bound $\epsilon(k)$ and the implicit bound $g(k)$ under different settings of N and β .

is performed for the number of scenarios, N , ranging from 1 to 100, and for $\beta = 0.01$. In each repetition, a set of N scenarios is sampled, the global minimax point (x_N^*, y_N^*) is computed, and its complexity s_N^* is recorded to determine the theoretical bound $\epsilon(s_N^*)$. Following this, the solution is subject to an out-of-sample test against a large independent set of $M_{\text{test}} = 10^4$ test scenarios. This test yields the empirical risk $\hat{V}_N(x_N^*, y_N^*) = \frac{1}{N_{\text{test}}} \sum_{i=1}^{M_{\text{test}}} \mathbb{1}\{(x_N^*, y_N^*) \notin \mathcal{X}_{\delta_i} \times \mathcal{Y}_{\delta_i}\}$, for that repetition. The average empirical risk over R repetitions in the Monte Carlo simulation is denoted by $\bar{V}_N(x_N^*, y_N^*)$.

Unlike the risk, the complexity s_N^* is identical across all Monte Carlo repetitions for any N . This is due to the special structure of this numerical example, specifically the bilinear payoff and interval uncertainty. Therefore, $\epsilon(s_N^*)$ in Fig. 4.2 remains constant for a given N . Moreover, we observe that the empirical risk can be exactly zero, implying the absence of a strictly positive

lower bound. By fixing $\beta = 0.01$ (a 99% confidence level), we expect the empirical risk $\hat{V}_N(x_N^*, y_N^*)$ to exceed the theoretical bound $\epsilon(s_N^*)$ in no more than 1% of the repetitions. In Fig. 4.2, our numerical results are consistent with this guarantee, as the bound was never violated, and also for the cases $\beta = 0.1$ or 0.0001. This provides a numerical verification of Theorem 6.

Finally, we set $N \in \{100, 1000\}$ and $\beta \in \{10^{-2}, 10^{-5}\}$. The comparisons between the explicit bound $\epsilon(k)$ (Theorem 5) and the implicit bound $g(k)$ (Theorem 6) for $k = 0, 1, \dots, N$ are presented in Figure 4.3.

As illustrated in Figure 4.3, we numerically verify the theoretical relationship between the explicit bound $\epsilon(k)$ and the implicit bound $g(k)$. Two primary phenomena can be distinctly observed across all four subplots. First, the implicit bound $g(k)$ (red dashed line) remains consistently lower than or equal to the explicit bound $\epsilon(k)$ (blue solid line) over the entire range of k . This geometrically confirms that $g(k)$ yields a tighter and less conservative generalization guarantee. Furthermore, the tighter bound $g(k)$ is very close the line k/N , especially when N is large. Second, the two curves perfectly intersect at $k = N - 1$ and $k = N$, corroborating our analytical derivations. Notably, these qualitative behaviors are invariant to the scaling of the sample size N and the confidence parameter β . This validates the theoretical result in Proposition 3.

Chapter 5

Conclusions and Future work

This chapter summarizes the theoretical findings of the thesis, discusses the limitations of this current framework established, and identifies possible directions for future work.

5.1 Conclusions

A data-driven and distribution-free framework was developed in this thesis to establish probabilistic robustness guarantees for an ε -stationary point and a global minimax point in general NC–NC minimax problems. In particular, a tight robustness guarantee for a local equilibrium was obtained through an ε -stationary point under convex constraint sets. Conversely, under fully non-convex constraint sets, it was demonstrated that global minimax points adhere to an alternative, relaxed robustness bound. A critical direction for future research was to theoretically establish a tight probabilistic bound for global minimax points, a possibility strongly hinted at by the numerical validation of the tight bound in Theorem 2. The extension of this framework to non-convex multi-player games was also identified as the most challenging direction.

We should note that, while this thesis successfully employs the scenario approach to address NC–NC minimax problems with uncertain constraints, it is important to acknowledge the inherent limitations of this methodology.

First, as pointed out by the existing literature [63, 64], even for convex problems, the required sample size scales linearly with the problem's dimensions to achieve robust probabilistic guarantees. For NC–NC minimax problems with uncertain constraints, these standard theoretical bounds do not directly apply. In non-convex settings, the required sample size is typically much higher—often requiring bounds based on the complexity s_N^* , rather

than simple parameter dimensions—which means an even larger sample size would theoretically be needed to ensure the same level of robustness. For instance, as demonstrated in Chapter 4, when fixing $\beta = 0.01$, even if $N = 100$ scenarios are drawn and the complexity s_N^* remains very small, the theoretical upper bound on the risk is still remarkably loose. This highlights a fundamental trade-off, echoing the well-known “no free lunch” principle: a purely data-driven approach that demands minimal prior knowledge must naturally compromise on the tightness of its theoretical bounds.

Second, solving non-convex minimax problems is already computationally expensive. Adding the need for a large sample size makes it difficult to apply this method to larger problems. In addition, the final solution is only reliable if the sampled data closely follows the true, unknown distribution.

5.2 Future Work

As previously discussed, the high sample complexity associated with the scenario approach introduces significant computational burdens. While the developed theoretical framework provides robust distribution-free guarantees, computing the actual equilibria over massive scenario sets remains an open challenge. Consequently, future research may address this computational issue by developing tailored algorithmic approaches. From a control-theoretic perspective, integral quadratic constraints [65] provide a powerful and unified framework for algorithm analysis, as successfully demonstrated in convex optimization [66, 67]. A natural question is whether, and how, this control-theoretic framework can be extended to analyze algorithms for non-convex optimization, and more importantly, the highly challenging non-convex minimax problems.

Beyond the algorithmic challenges discussed above, another fundamental direction lies in expanding the theoretical scope of the studied equilibria. The ε -stationary point and the global minimax point are two key classes of equilibria in NC–NC minimax problems; however, numerous other definitions of equilibria exist in the literature. For example, [54] introduced the local minimax point, revealing a highly counterintuitive fact: a global minimax point may be neither a local minimax nor a stationary point [54, Prop. 21]. This stands in contrast to the basic property of (non-convex) minimization problems, where global minima are always local minima. As another example, [68] proposed an approximate local min-max equilibrium, termed the (ε, δ) -local min-max solution [68, Def. 1.1]. Such points possess desirable properties regarding their existence, computational complexity,

and connection to GDA algorithms. While there are certainly many other equilibrium concepts not covered in this thesis, investigating the robustness of these various equilibria remains a topic of great interest and importance.

A natural extension of this thesis is to generalize the current zero-sum framework to the setting of non-convex multi-player games. For example, [56] investigated a specific class of these games wherein the players' payoff functions are composed of quadratic operators and canonical functions. Although the existence of a global NE is theoretically guaranteed in this formulation, the (probabilistic) robustness of these equilibria represents an open and relatively unexplored area. More broadly, while several studies have investigated various non-convex game formulations (e.g., [69, 70, 71]), establishing probabilistic robustness guarantees within these complex settings remains an important open challenge.

Finally, recent progress in non-convex scenario optimization, especially the scenario-based decision-making framework introduced by Prof. Simone Garatti and Prof. Marco Campi (discussed in Section 2.1.3), has paved the way for addressing many difficult problems not limited to optimization. However, this current framework is not applicable to the global minimax point in NC–NC minimax problems, as demonstrated in Section 3.3. Hence, developing a tailored framework specifically for minimax problems and broader game-theoretic contexts represents a compelling direction for future research.

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